

RESIDUAL-GATED DUAL ATTENTION FOR REAL-TIME POTHOLE DETECTION

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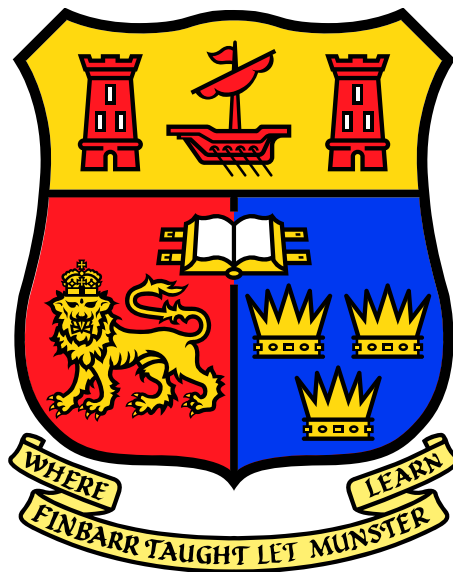
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Abstract

Potholes are hard to localise automatically: irregular in geometry, weakly bounded, highly variable in apparent scale, and easily confused with shadows, patched asphalt and wet road surfaces. This dissertation presents RGDA-YOLOv8m, which inserts Efficient Channel Attention followed by the Convolutional Block Attention Module at three multiscale feature-fusion stages of the YOLOv8m neck. The modules are combined with the unmodified features through a learnable scalar gate initialised to zero, so the network begins fine-tuning as an exact copy of the pretrained detector. Evaluation uses a deduplicated corpus of 926 images and 2,354 annotations built from three public Kaggle repositories, from which 1,078 duplicates were removed by normalized-pixel nearest-neighbor matching, and compares nine configurations. Across three seeds RGDA-YOLOv8m attains an mAP@0.5 of 0.8141 ± 0.0142 and an F1 of 0.7708 ± 0.0121 , exceeding the identically trained no-attention baseline in every paired run, for 0.25% additional parameters and 8.1% of throughput, at 139.9 frames per second. Ablation shows that neither module helps alone. Only their combination does. Applying the same block to YOLOv11m, whose backbone already contains partial self-attention, instead degrades both metrics. The value of an added attention module is therefore a property of the pairing, not of the module.

Declaration

I confirm that, except where indicated through the proper use of citations and references, this is my original work and that I have not submitted it for any other course or degree.

Signed: _____

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August 30, 2026

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Chapter 1

Introduction

1.1 Background and Motivation

Road-surface deterioration is one of the most persistent and least glamorous problems in transport infrastructure. Potholes form where water penetrates the pavement structure, weakens the sub-base, and is then worked loose by repeated traffic loading and freeze–thaw cycling. The resulting cavities damage suspensions, wheels and tyres, force sudden avoidance manoeuvres, and impose substantial repair and compensation costs on the authorities responsible for the network [Eriksson et al. 2008; Wei Wang et al. 2020]. The cost is recurrent rather than one-off: a pothole that is not repaired grows, and the cost of repair grows with it, so the economic case for road maintenance rests almost entirely on detecting defects early.

Detection, however, is the bottleneck. The conventional approach is manual survey, in which trained inspectors either walk or drive the network and record defects by eye. Manual survey is accurate when it is done carefully, but it is slow, expensive, hazardous on live carriageways, and inconsistent between inspectors [N. Ma et al. 2022]. A road authority responsible for thousands of kilometres of network cannot survey all of it frequently enough to catch defects while they are still cheap to repair. This mismatch between the rate at which defects appear and the rate at which they can be found is the practical motivation for automating the detection step.

The safety consequences extend beyond vehicles. Surface defects on footpaths, at pedestrian crossings and along walking routes are a trip hazard, and the hazard falls disproportionately on people who cannot see the defect in advance. An estimated 43.3 million people worldwide are blind and a further 295 million experience moderate-to-severe distance vision impairment [GBD 2019 Blindness and Vision Impairment Collaborators 2021]. The World Health Organization places the total number of people living with some degree of vision impairment at at least 2.2 billion, including at least 1 billion cases that are preventable or unaddressed [World Health Organization 2024]. For this population an automated detector that runs on a phone or a wearable camera and warns of an approaching surface defect is not a convenience but an accessibility

technology, and several recent systems have been built with exactly this application in mind [Paramarthalingam et al. 2024]. A detector intended for that role has a hard latency requirement: a warning that arrives after the user has already stepped into the defect is worthless.

These two applications, network-scale maintenance survey and assistive navigation, share a common technical requirement. Both need a detector that processes a video stream in real time on modest hardware, and both are sensitive to the two ways a detector can fail. A missed pothole is a hazard that goes unreported. A false alarm on a shadow or a tar patch erodes trust in the system and, in the maintenance case, wastes an inspection visit. Neither precision nor recall alone captures what is wanted, which is why the balanced F1 score and mean average precision are both reported throughout this dissertation.

1.1.1 What Makes Pothole Detection Difficult

Potholes lack the three properties that make general-purpose object detection tractable: a canonical appearance, a characteristic shape, and a boundary that separates figure from ground.

A pothole has no canonical form. Defined by the *absence* of pavement, its outline follows how the material failed rather than any shape prior, and two potholes a metre apart may share no visual property beyond being darker than their surroundings, and sometimes not even that.

The boundary is frequently weak. Deterioration is progressive, so the transition from intact pavement to open cavity is gradual and the point at which an annotator draws the box is partly a judgement, which bounds achievable localisation accuracy by annotation consistency rather than by the detector.

The confounders share the defect’s discriminating features. Cast shadows, tar patches, wet regions, oil stains, drainage grates, manhole covers and worn lane markings all produce dark, irregular or coarsely textured road regions, so the task is fine-grained discrimination rather than coarse object recognition.

The apparent scale varies enormously. A pothole ten metres ahead fills a substantial fraction of the frame. At sixty metres it occupies a handful of pixels. Useful warning requires detection at range, which is what makes the multi-scale fusion stage targeted by this dissertation central to the task.

1.2 Research Gap and Objectives

Three specific gaps motivate the work reported here.

First, controlled cross-architecture comparison is rare. Published pothole-detection results are reported on different datasets, with different partitions, different preprocessing, different average-precision definitions and different hardware. Two papers reporting mAP@0.5 for the same nominal architecture routinely differ by more than the improvements they claim, and there is usually no way to determine from the

papers whether the difference is architectural or procedural. Relatively few studies place one-stage, two-stage, transformer-based, attention-enhanced and ensemble detectors on a single deduplicated dataset under a single evaluation protocol.

Second, the architecture dependence of attention is untested. The effectiveness of an added attention module is normally reported as though it were a property of the module. Whether that effectiveness survives a change of host (specifically, a host that already contains a spatial self-attention stage) has not been systematically examined for this task.

Third, single-run reporting is the norm. Most pothole-detection studies report the outcome of one training run. Deep detectors trained on small datasets have substantial run-to-run variance from initialisation and data ordering, and an improvement smaller than that variance is not evidence of anything. Without repeated runs there is no way for a reader to tell which category a reported improvement falls into.

The objectives of this dissertation follow directly. It compares representative one-stage, two-stage, transformer-based, attention-enhanced and ensemble detectors on a single deduplicated pothole dataset under one fully specified evaluation protocol. It introduces a parameter-light dual-attention block, combining channel and spatial recalibration and gated so that it does not disturb the pretrained representation at the start of fine-tuning, and measures its effect on the accuracy and throughput of YOLOv8m. It isolates the individual and joint contributions of the block's two pathways. And it tests whether any benefit observed on YOLOv8m transfers to YOLOv11m, whose backbone already contains a partial self-attention stage.

Figure 1.1 summarises the framework through which these objectives are addressed: a ten-configuration benchmark on the deduplicated dataset, the residual-gated ECA-CBAM integration at three YOLOv8m neck stages, an analysis of architecture-dependent attention behaviour, and an evaluation of the resulting accuracy-throughput trade-off.

1.3 Contributions

The contributions of this dissertation are as follows. Throughout, detection accuracy is reported as mean average precision at an intersection-over-union threshold of 0.5, written mAP@0.5, and throughput is reported in frames per second (FPS).

- **A deduplicated multi-source pothole dataset and the procedure that produced it.** Three public Kaggle repositories were consolidated and then deduplicated by normalized-pixel nearest-neighbor matching, which removed 1,078 of 2,004 images, more than half the corpus, because the repositories redistribute overlapping data. Byte-level hashing was tried first and failed, for reasons documented in Chapter 4. The resulting corpus contains 926 images and 2,354 annotations.
- **A controlled ten-configuration benchmark.** RGDA-YOLOv8m and nine comparators (YOLOv8m, YOLOv10m, YOLOv11m, Faster R-CNN, SSD-VGG16, RT-DETR, a YOLOv8m and Faster R-CNN ensemble combined by Weighted Box

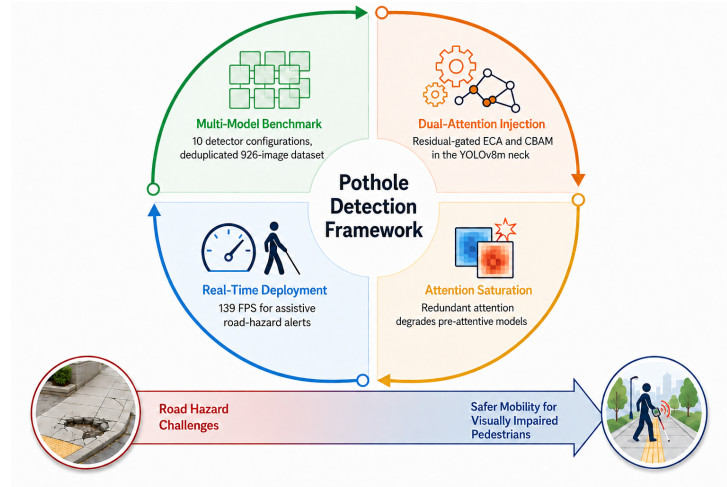


Figure 1.1: Overview of the RGDA-YOLOv8m pothole detection framework: dataset consolidation and deduplication, the residual-gated dual-attention architecture, the cross-architecture benchmark, and the ablation and architecture-dependence analyses.

Fusion, YOLOv8m+CBAM and YOLOv8m+CoordAttn) are evaluated on the same 186-image validation partition under one average-precision definition and one throughput-measurement procedure.

- **The RGDA-YOLOv8m architecture.** Sequential ECA and CBAM refinement is applied at three YOLOv8m neck stages and combined with the unrefined features through a learnable scalar gate initialised to zero, so the network at initialisation is exactly the pretrained detector. Across three seeds the model attains an mAP@0.5 of 0.8141 ± 0.0142 and an F1 of 0.7708 ± 0.0121 , improvements of 0.82 and 1.07 percentage points over the identically trained baseline, for 64,824 additional parameters (0.25%) and 139.9 FPS.
- **A factorial ablation of the two attention pathways.** The no-attention, ECA-only, CBAM-only and combined conditions are compared under an identical training schedule across three seeds. Neither single module improves on the baseline. Only the combination does.
- **The attention-saturation result.** Attaching the identical block to YOLOv11m reduces both mAP@0.5 and F1 at every seed. The comparison is against a single-run baseline and so is weaker than the three-seed YOLOv8m ablation, but it is the result with the broadest implication: the benefit of an added attention module may be a property of the pairing rather than of the module.
- **An account of protocol sensitivity.** Three methodological choices (deduplication, the average-precision definition, and the throughput-measurement method) are

each shown to move the reported numbers by more than the architectural effect under investigation.

1.4 Thesis Structure

Chapter 2 reviews classical, CNN-based, YOLO-family, attention-enhanced and ensemble approaches to pothole and road-damage detection, together with the datasets and evaluation practices of the field. Chapter 3 describes the detector architectures used in the benchmark and the software framework within which they were extended. Chapter 4 documents the construction, deduplication and statistical properties of the dataset. Chapter 5 presents the design of the residual-gated dual-attention block, its integration by forward hooks, and the two-phase training protocol. Chapter 6 specifies the evaluation metrics and the measurement procedures used to obtain them. Chapter 7 reports and discusses the benchmark, the multi-seed evaluation, the ablation, the architecture-dependence experiment and the speed–accuracy trade-off. Chapter 8 draws the conclusions, states the limitations and sets out directions for further work.

Chapter 2

Related Work

This chapter reviews the literature that positions the work reported here: classical image processing and non-visual sensing; the transition to convolutional detection; YOLO-family architectures adapted for road defects; attention mechanisms in detection architectures; transformer-based detection; ensemble and fusion approaches; and the datasets and evaluation practices that determine whether any of these results can be compared with one another. It closes by stating the gaps that the work reported here addresses.

2.1 Classical and Sensor-Based Pothole Detection

The earliest vision-based detectors were built from explicit rules about what a pothole looks like: Koch and Brilakis [Koch, and Brilakis 2011] segment a pavement image by histogram-shape-based thresholding, approximate each candidate's shape by morphological thinning and elliptic regression, and accept or reject it by comparing interior and surrounding texture, while related work applied edge detectors and the Hough transform to video [Nienaber, Booysen, and Kroon 2015]. Such methods need no training data and fail legibly, but their assumptions are fixed at design time while the confounders listed in Section 1.1.1 are not, so they grow brittle in proportion to the special cases they accumulate. Vision is not the only modality: Pothole Patrol [Eriksson et al. 2008] inferred anomalies from the vertical acceleration of taxi suspensions, turning a fleet into a survey instrument at almost no marginal cost, but a vibration sensor reports only defects the vehicle drives over and only after contact. Three-dimensional methods instead measure the surface geometrically, and Fan et al. [Fan et al. 2022] show that fitting a reference plane to a reconstruction isolates depressions in a way largely immune to appearance confounds. Ma et al. [N. Ma et al. 2022] organise the field along this axis and note that the sensing hardware 3-D reconstruction requires is what has kept it out of consumer-scale deployment.

2.2 Convolutional Detection for Road Damage

Convolutional methods entered this domain as patch classification, when Zhang et al. [Lei Zhang et al. 2016] showed a learned representation outperforming handcrafted descriptors on the same data. Yang et al. [F. Yang et al. 2020] extended this to pixel-level segmentation and Li et al. [Shanchuan Li, Cao, and Cai 2023] added uncertainty estimation, segmentation suiting quantification of a defect's area and detection suiting warning or triage, as here. Detection proper was driven by the road-damage challenges: Maeda et al. [Maeda et al. 2018] trained an SSD-based detector on municipal smart-phone imagery, showing a single-stage detector on commodity hardware could produce usable survey data, and Wang et al. [Wei Wang et al. 2020] found cross-city generalisation markedly worse than within-city, an early sign of location-specific appearance bias. The two-stage lineage from R-CNN [Girshick et al. 2014] to Faster R-CNN [Ren et al. 2015] localises well but at substantial cost, appearing here mainly as an ensemble component. Al-Shaghouri et al. [Al-Shaghouri, Alkhatib, and Berjaoui 2021] give the most careful pothole-specific comparison, evaluating SSD, YOLOv3 and YOLOv4 on 1,087 images under matched conditions, YOLOv4 reaching 85.39% mAP at roughly 20 FPS with reliable detection to one hundred metres, a range that determines how much warning time the user gets.

2.3 YOLO-Family Architectures for Road Defect Detection

The YOLO family reformulated detection as a single regression from image to boxes and class scores [Redmon et al. 2016], which is what makes it fast enough for video. YOLOv8 [Ultralytics 2023] pairs C2f blocks, built on the residual-learning principle of K. He et al. [2016], with a PAN-FPN neck and an anchor-free decoupled head, addressing the foreground-background imbalance of dense prediction by task-aligned assignment rather than focal loss [Lin, Goyal, et al. 2017]. YOLOv10 [A. Wang et al. 2024] added dual assignments so that non-maximum suppression can be dropped at inference; and YOLO11 [Ultralytics 2024] replaced C2f with C3k2 blocks and, critically here, inserted a C2PSA partial-self-attention module after the backbone's spatial pyramid pooling stage, each version embedding more attention-like machinery in the base architecture. Much later work adapts them for constrained hardware or multi-scale fusion [X. Li, and Y. Zhang 2024; Q. He, Z. Li, and W. Yang 2024; Pei, Wu, and X. Liu 2024]. Two findings bear directly on this study: LRS-YOLO gains 7.2 percentage points of mAP@50 on RDD2022 but only 2.8 on UAV-PDD2023 [Chen, Yu, and Shu Li 2024], the dataset-side analogue of the architecture-side dependence examined here, and Pham et al. [Pham, Ngoc, and Bui 2024] report that newer YOLO versions did not uniformly outperform older ones once inference time was accounted for, as Chapter 7 also finds. For scale, Abdelwahed et al. [Abdelwahed et al. 2025] benchmark YOLOv10m, v11m and v12m at 0.511, 0.515 and 0.525 mAP@0.5:0.95 and 223, 213 and 206 FPS.

2.4 Attention Mechanisms in Detection Architectures

Channel attention began with Squeeze-and-Excitation [Hu, Shen, and Sun 2018], which pools each channel to a scalar and rescales it but forces every channel interaction through a bottleneck. Efficient Channel Attention [Q. Wang et al. 2020] replaced that bottleneck with a small one-dimensional convolution, Coordinate Attention [Hou, Zhou, and Feng 2021] factorised the pooling to retain positional structure, the Convolutional Block Attention Module [Woo et al. 2018] added a spatial gate after a channel gate, BAM [Park et al. 2018] arranged the two in parallel, and non-local networks [Xiaolong Wang et al. 2018] generalised spatial attention to explicit long-range interaction. ECA and CBAM are specified in full in Chapter 5. Results in road-damage papers are almost uniformly positive [Pei, Wu, and X. Liu 2024; Chen, Yu, and Shu Li 2024; Pham, Ngoc, and Bui 2024], two being directly relevant: YOLO9tr [Youwai, Chaiphaphat, and Chaipetch 2024] found a single-attention-step variant underperforming its full configuration, and Zhang et al. [Linxuan Zhang, Deng, and Zou 2025] add attention beneficially to a YOLOv11 host, raising mAP@0.50 from 88.2% to 91.2%, the opposite of what Chapter 7 observes and a tension taken up in Section 7.8.2. Interpretation is complicated by attention rarely being introduced alone [Chen, Yu, and Shu Li 2024; Pei, Wu, and X. Liu 2024; Linxuan Zhang, Deng, and Zou 2025; Lu et al. 2025] and by single-run ablations that confound it with run-to-run variation. Underneath lies an unexamined assumption: attaching a module to one host estimates an effect size only if that effect is host-independent. No study located here tests that by holding the block and protocol fixed and varying only the host.

2.5 Transformer-Based Detection

DETR [Carion et al. 2020] reformulated detection as direct set prediction over CNN features with a bipartite matching loss, removing both anchors and non-maximum suppression, and the self-attention it inherits from the transformer architecture [Vaswani et al. 2017] gives every feature position access to every other, which is qualitatively different from the local reweighting performed by modules such as CBAM. Its practical weaknesses were slow convergence and poor small-object performance. RT-DETR [Zhao et al. 2023] addressed the speed problem with a hybrid encoder decoupling intra-scale interaction from cross-scale fusion, achieving throughput competitive with YOLO detectors while retaining NMS-free set prediction, and is this dissertation's representative of that paradigm. As Chapter 7 shows, its confidence calibration differs from that of NMS-based detectors in a way that has direct consequences for how detectors should be compared. The Swin Transformer [Z. Liu et al. 2021] made transformer backbones practical for dense prediction and appears in road-damage systems such as the ensemble of Ding et al. [Ding et al. 2022]. The relevance here is conceptual, since Swin, C2PSA and the partial attention block of YOLO9tr all place self-attention inside the feature extractor, precisely the condition under which Chapter 7 suggests further external attention

becomes counterproductive.

2.6 Ensemble and Fusion Approaches

Combining detectors requires deciding what to do with overlapping boxes: non-maximum suppression discards every box overlapping a higher-scoring one beyond a threshold and Soft-NMS [Bodla et al. 2017] decays their scores instead, but both are suppression rules, so the surviving box is one of the inputs and the information in the discarded boxes is lost, whereas Weighted Box Fusion [Solovyev, Weimin Wang, and Gabruseva 2021] clusters overlapping detections and computes a fused box whose coordinates are a confidence-weighted average of the cluster members, exploiting the partly independent localisation errors of independent detectors at the cost of running every member model. Ding et al. [Ding et al. 2022] built an ensemble of one-stage, two-stage and Swin-Transformer-based detectors for the CRDDC2022 challenge and report a mean F1 of 0.7699 across three country subsets, a representative result in that ensembling reliably improves accuracy on road-damage benchmarks and reliably destroys real-time throughput. This dissertation includes a YOLOv8m and Faster R-CNN ensemble combined by Weighted Box Fusion specifically to quantify that exchange on the same data as the single-model configurations.

2.7 Datasets, Evaluation and Comparability

Road-damage research is organised around a few public corpora: RDD2022 [Arya et al. 2022] underpins the CRDDC challenges. UAV-PDD2023 supplies aerial imagery on which Pan et al. [Pan, Xu Wang, and Huan 2025] report an mAP@0.5 of 0.542, a reminder that reported accuracy is dominated by task difficulty rather than architecture quality. Pothole-specific work relies on Kaggle repositories [Chitholian 2020; Larxel (AndrewMVD) 2020; Ashish Kumar 2021] that are convenient, small and, as Chapter 4 shows, substantially overlapping; and robustness-oriented corpora are rare, Frnda et al. [Frnda et al. 2024] being an exception in spanning clear, rainy, night, evening and sunset conditions and finding accuracy degrading sharply under adverse illumination. The overlap matters, because a re-encoded copy of an image appearing in both partitions inflates every metric invisibly, and very few studies report a deduplication step. Nor is average precision a single quantity: the VOC 2007 rule averages interpolated precision at eleven recall levels while COCO integrates over 101 [Everingham et al. 2010; Lin, Maire, et al. 2014], giving different numbers from identical predictions, and frames per second is worse still. Compounding this, most studies report one run without any estimate of its variance, so no later reader can tell whether two results are consistent. The three-seed protocol adopted here is a minimal response.

2.8 Synthesis and Research Gaps

Table 2.1 summarises the studies most directly relevant to this dissertation and states what each contributes to it.

Table 2.1: Studies most directly relevant to this dissertation and the role each plays in it.

Study	Approach	Relevance to this work
Koch, and Brilakis [2011]	Histogram thresholding, morphology, texture comparison	Establishes the appearance assumptions that learned detectors remove
Eriksson et al. [2008]	Accelerometer and GPS sensing	Contrasting modality; motivates forward-looking vision
N. Ma et al. [2022]	Review of road imaging and pothole detection	Organises the sensing and algorithmic design space
Maeda et al. [2018]	SSD on municipal smartphone imagery	Demonstrates single-stage feasibility at survey scale
Al-Shaghouri, Alkhatib, and Berjaoui [2021]	SSD, YOLOv3, YOLOv4 compared on 1,087 pothole images	Matched-protocol comparison; detection range matters for warning
Ren et al. [2015]	Faster R-CNN with region proposal network	Supplies the two-stage comparator and the ensemble member
Zhao et al. [2023]	Real-time DETR with hybrid encoder	Supplies the set-prediction comparator; motivates dual operating points
Q. Wang et al. [2020]	Efficient Channel Attention	Channel-recalibration component of the proposed block
Woo et al. [2018]	Sequential channel and spatial attention	Channel-spatial component of the proposed block
Hou, Zhou, and Feng [2021]	Direction-aware channel attention	Alternative attention comparator in the benchmark
Pei, Wu, and X. Liu [2024]	YOLOv8 with DSC-C2f and coordinate attention	Representative positive attention result on a YOLOv8 host

Continued on next page

Table 2.1 (continued from previous page)

Study	Approach	Relevance to this work
Chen, Yu, and Shu Li [2024]	YOLOv8n with RFACnv, SPPFCSPPC and LKCA	Shows the same modification giving very different gains per dataset
Linxuan Zhang, Deng, and Zou [2025]	Improved YOLOv11 with fusion attention	Reports attention gains on a YOLOv11 host; contrasted in Chapter 7
Youwai, Chaiyaphat, and Chaipetch [2024]	YOLOv9 with a partial attention block	Reports that the number of attention stages matters independently
Ding et al. [2022]	One-stage and two-stage ensemble for CRDDC2022	Establishes the ensemble accuracy–throughput exchange
Solovyev, Weimin Wang, and Gabruseva [2021]	Weighted Box Fusion	Fusion rule used to build the ensemble comparator
Frnda et al. [2024]	YOLOv7 and Faster R-CNN under five weather conditions	Identifies the robustness limitation of fair-weather corpora
Paramarthalingam et al. [2024]	YOLOv5 for assistive pothole detection	Closest application match; external accuracy reference point

Three gaps follow from this review, and they are what the objectives of Section 1.2 address. First, no located study places one-stage, two-stage, transformer-based, attention-enhanced and ensemble detectors on a single deduplicated pothole dataset under one fully specified protocol. Second, no located study isolates the individual and joint contributions of a channel and a spatial attention pathway inserted at the same locations under an identical schedule. Third, and most importantly, no located study holds the attention block and training protocol fixed and varies only the host architecture in order to test whether the benefit of attention is a property of the module or of the pairing.

Chapter 3

Detector Architectures and Experimental Framework

This chapter describes the detection architectures evaluated in this dissertation and the software framework within which they were trained, extended and measured. Its purpose is to make the later chapters legible: the design of the proposed attention block in Chapter 5 depends on specific structural properties of the YOLOv8m neck, and the architecture-dependence result in Chapter 7 depends on a specific structural difference between YOLOv8m and YOLOv11m. Both are set out here.

3.1 Anatomy of a Modern Detector

3.1.1 Backbone, Neck and Head

Contemporary convolutional detectors share a three-part decomposition. The *backbone* is a convolutional feature extractor that consumes the input image and produces a pyramid of feature maps at progressively lower spatial resolution and higher channel dimension. The stages of that pyramid are conventionally labelled P1 to P5, where P_n has been downsampled by a factor of 2^n . For a 640×640 input the P3, P4 and P5 stages have spatial resolutions of 80×80 , 40×40 and 20×20 respectively. Shallow stages retain spatial detail but encode weak semantics. Deep stages encode strong semantics but have lost the resolution needed to localise small objects.

The *neck* exists to resolve that tension. A Feature Pyramid Network [Lin, Dollár, et al. 2017] propagates semantic information downward from deep stages to shallow ones through a top-down path with lateral connections, so that the high-resolution features used to detect small objects are semantically informed. PANet [S. Liu et al. 2018] adds a second, bottom-up path so that the precise localisation signal available in shallow layers also propagates upward. The combination, referred to as PAN-FPN, is the neck design used by YOLOv8 and its immediate successors.

The *head* converts each neck output into predictions. A detection is produced at

each spatial position of each pyramid level, so the three levels naturally specialise to small, medium and large objects.

3.1.2 Anchor-Free Prediction and Label Assignment

Older detectors predicted offsets relative to a fixed set of anchor boxes whose sizes and aspect ratios had to be chosen in advance, often by clustering the training annotations. Anchor-free detectors such as FCOS [Tian et al. 2019] instead regress the distances from each position to the four sides of the box directly, which removes a set of dataset-specific hyperparameters. YOLOv8 is anchor-free and uses a distribution-focal formulation in which each box side is predicted as a discrete distribution over candidate offsets rather than as a single scalar.

Label assignment, deciding which predicted positions are responsible for which ground-truth boxes, is handled by a task-aligned assigner that scores candidate positions by a combination of classification confidence and localisation quality, rather than by the fixed geometric rules used in earlier detectors. This matters for pothole detection because potholes have irregular outlines that fill their bounding boxes poorly, so a purely geometric assignment rule tends to select positions that lie on the road surface rather than on the defect.

3.1.3 Receptive Field and Small-Object Detection

The reason multi-scale fusion is not optional for this task can be stated in terms of receptive fields. A unit in the P5 stage of a YOLOv8 backbone responds to a region of the input image spanning a large fraction of the frame, which is what allows it to encode context, that this dark region sits in a road rather than in foliage. A unit in P3 responds to a much smaller region and can therefore resolve a defect only tens of pixels across, but has correspondingly little context with which to distinguish it from a shadow.

Neither stage alone is adequate for potholes, because the same physical defect appears at both scales depending on its distance from the camera. A pothole ten metres ahead is a P5-scale object. The same pothole at sixty metres is a P3-scale object, and it is the distant one whose detection provides useful warning time. The corpus described in Chapter 4 contains instances across this entire range within single images, since a road scene recedes towards a vanishing point and defects at several distances are simultaneously visible.

The consequence for this dissertation's design is direct. If the detector's difficulty is concentrated in reconciling scales, then the fusion stage is where a content-dependent reweighting has the most to work with: it is the only place in the network where features carrying fine spatial detail and features carrying scene context are present in the same tensor. This is the architectural argument for the injection points chosen in Chapter 5, complementing the empirical argument from the dataset properties given in Section 4.5.

3.1.4 Model Scaling and the Choice of the Medium Variant

Ultralytics architectures are published in five sizes (n, s, m, l and x) generated from one topology by two multipliers, one scaling channel width and one scaling block repetition, so the variants differ only in width and depth. The medium variant was used for every YOLO configuration in this dissertation. The nano and small variants are the natural choice for embedded deployment and dominate the lightweight-modification literature of Chapter 2, but they leave little headroom, so a 0.25 percent parameter addition sits closer to the noise floor of what the architecture can express. The large and extra-large variants would reduce throughput below the real-time threshold that motivates the work. At approximately 25.9 million parameters and over 150 FPS on the target hardware, the medium variant is the smallest configuration for which an 8.1 percent throughput cost is clearly affordable and the largest for which real-time operation is uncontroversial. Holding the scale fixed throughout is also a control: comparing an attention-augmented small model against an unaugmented medium one would confound the mechanism with capacity.

3.1.5 The Role of Feature Recalibration

Neither the backbone nor the neck contains any mechanism that explicitly decides which channels or which spatial locations of a fused feature map are informative. Convolution is content-agnostic in the sense that its weights are fixed once training is complete and do not depend on the particular input. Attention modules supply exactly this input-dependent reweighting, and the neck is the natural place to apply it: it is where multi-scale information has been combined but before predictions are formed, so a recalibration applied there influences every subsequent prediction at that scale. This observation motivates the injection points chosen in Chapter 5.

3.2 YOLOv8m: The Host Architecture

3.2.1 C2f Feature Extraction

The YOLOv8 backbone is built from C2f blocks [Ultralytics 2023]. A C2f block splits its input along the channel dimension, routes one part through a sequence of bottleneck units, and concatenates the outputs of every bottleneck with the untransformed part before a final projection. Compared with the C3 block it replaces, C2f exposes more gradient paths for the same parameter budget, which improves optimisation without increasing inference cost proportionally. The backbone terminates in a spatial pyramid pooling – fast (SPPF) module that applies successive max-pooling operations with a shared kernel and concatenates the results, giving the deepest stage a large effective receptive field at low cost.

3.2.2 Inside the C2f Block

Because the injection points of Chapter 5 are C2f block outputs, the block's internal structure determines what those outputs contain. A C2f block first applies a 1×1 convolution that expands the input to twice a working width, then splits the result into two halves along the channel dimension. One half is passed through unchanged. The other is fed through a chain of n bottleneck units, each consisting of two 3×3 convolutions with an optional identity shortcut, and the output of *every* unit in the chain is retained. All retained tensors, the untouched half and the n intermediate outputs, are concatenated and projected back to the target width by a final 1×1 convolution.

Two properties follow. First, the block's output is a concatenation of representations computed at different effective depths, so its channels are heterogeneous by construction: some describe features that have passed through one bottleneck, others through several, and one group through none at all. Channel attention applied to such a tensor is therefore not merely selecting among redundant filters but selecting among representations of genuinely different character, which is a plausible reason for it to be useful here. Second, retaining every intermediate output creates multiple short gradient paths from the block output back to its input, which is the optimisation benefit C2f was introduced to obtain.

The backbone closes with an SPPF module, which applies the same 5×5 max-pooling operation three times in succession and concatenates the input with all three pooled results. Chaining a single kernel reproduces the effective receptive fields of larger kernels at a fraction of the cost, and the concatenation gives the deepest stage a representation that mixes several context scales. In YOLO11 this is exactly the point at which the C2PSA self-attention module is inserted, which is why the difference between the two hosts is a difference in what the *deepest* features have already undergone.

3.2.3 The PAN-FPN Neck

The YOLOv8 neck implements the PAN-FPN pattern described above. The top-down path upsamples P5, concatenates it with P4 and processes the result through a C2f block. The output is upsampled again, concatenated with P3 and processed through a further C2f block, yielding the highest-resolution neck output. The bottom-up path then downsamples that output, concatenates it with the intermediate top-down result and processes it, and repeats once more against P5. The three C2f blocks that terminate these fusion paths are the outputs delivered to the head.

For the medium-sized YOLOv8m configuration used throughout this dissertation, these three blocks emit feature maps with 192, 384 and 576 channels respectively. These are the three tensors that the attention block of Chapter 5 intercepts, and the channel dimensions determine the parameter cost of the injected modules.

3.2.4 The Decoupled Head

YOLOv8 uses a decoupled head: classification and box regression are computed by separate convolutional branches rather than from a shared output tensor. The two tasks require different features (classification benefits from translation invariance, localisation from translation equivariance), and separating them measurably improves both. There is no objectness branch. Classification confidence serves directly as the detection score.

3.2.5 Rationale for YOLOv8m as the Host

Three considerations led to YOLOv8m being selected as the host architecture for the proposed modification. First, it contains no attention mechanism of its own, which makes it a clean substrate: any effect observed can be attributed to the injected modules rather than to an interaction with an existing one. Second, the medium configuration at approximately 25.9 million parameters occupies the region of the accuracy–throughput curve relevant to real-time deployment on desktop-class hardware, so a throughput cost measured against it is meaningful. Third, it is the most widely used baseline in the recent road-damage literature reviewed in Chapter 2, which makes the results interpretable against that literature.

3.3 The Training Objective

The loss function is specified here because it is common to every YOLO-family configuration in this dissertation, including the proposed model, and because the ablation of Chapter 5 holds it fixed. YOLOv8 optimises a weighted sum of three terms computed over the positions selected by the task-aligned assigner:

$$\mathcal{L} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \lambda_{\text{dfl}} \mathcal{L}_{\text{dfl}}. \quad (3.1)$$

The classification term $\mathcal{L}_{\text{cls}}$ is binary cross-entropy over class logits. With a single class it reduces to a defect-versus-background decision at each selected position, which is substantially easier than the multi-class discrimination required by the road-damage benchmarks of Chapter 2 and is one reason absolute mAP values here are not comparable with those reported on RDD2022.

The localisation term $\mathcal{L}_{\text{box}}$ is a complete-IoU loss, which augments the plain IoU objective with penalties on the distance between box centres and on aspect-ratio mismatch. This matters for potholes specifically: an IoU-only objective provides no gradient when the predicted and ground-truth boxes are disjoint, which is a common early-training condition for small, low-contrast objects, whereas the centre-distance term remains informative there.

The distribution-focal term $\mathcal{L}_{\text{dfl}}$ supervises the discretised box representation described in Section 3.1. Because each box side is predicted as a distribution over candidate offsets rather than as a scalar, the loss encourages probability mass to concentrate on the two bins adjacent to the true offset. The practical effect is that the model can

express uncertainty about a boundary rather than being forced to commit to a point estimate, which is appropriate for potholes, whose edges are genuinely ambiguous. Several of the road-damage detectors reviewed in Chapter 2 modify this term, substituting shape-aware IoU variants [Lu et al. 2025; Gao, Y. Wang, and Z. Ma 2025]. No such modification is made here, precisely so that the effect attributed to the attention block is not confounded with a change of objective.

The loss weights, the assigner and the augmentation policy were left at their Ultralytics defaults for every configuration. This is a deliberate constraint rather than an omission: tuning them per configuration would improve absolute results while destroying the controlled comparison that the ablation and the architecture-dependence experiment depend on.

3.4 Contemporary YOLO Variants

YOLOv10 [A. Wang et al. 2024] is included as the throughput-oriented member of the family. It trains with consistent dual label assignments, a one-to-many branch supplying rich supervision alongside a one-to-one branch producing a single prediction per object, so that non-maximum suppression can be removed at inference without the accuracy loss a naive one-to-one assignment would incur.

YOLO11 [Ultralytics 2024] revises feature extraction, replacing C2f with C3k2 blocks that use smaller kernels in their bottleneck units, and introduces a C2PSA module positioned immediately after the SPPF stage at the end of the backbone. C2PSA applies *partial self-attention*: the input is split along channels, one part is processed through a self-attention block with an accompanying feed-forward network, and the result is concatenated with the untouched remainder. Self-attention here operates over spatial positions, so each position of the deepest feature map is reweighted according to its relationship with every other position.

The architectural consequence is central to this dissertation. In YOLOv8m, the features arriving at the neck have never been reweighted by any content-dependent mechanism. In YOLOv11m, they have: the deepest backbone stage has already undergone a round of spatial self-attention before the neck begins fusing scales. If the value of an added attention module lies in performing a recalibration the host has left undone, then the same module attached to these two hosts should not behave the same way. This is precisely the comparison performed in Section 7.4.

3.5 Non-YOLO Comparators

3.5.1 Faster R-CNN

Faster R-CNN [Ren et al. 2015] is the two-stage representative. A region proposal network slides over the backbone feature map and emits class-agnostic object proposals with objectness scores. A second stage pools features from each proposal to a fixed size and

classifies and refines it. The configuration used here is the ResNet-50-FPN-v2 variant with COCO-pretrained weights. Two-stage detection generally localises well because the second stage sees features aligned to a hypothesised object rather than to a fixed grid, at the cost of running that stage once per proposal.

3.5.2 SSD-VGG16

SSD [W. Liu et al. 2016] predicts boxes directly from multiple feature-map scales using fixed anchor boxes, with no proposal stage and no feature-pyramid fusion. It is included as a high-throughput, architecturally simple lower bound. It is used at its native 300×300 input resolution with a VGG16 backbone, which is a genuine disadvantage relative to the 640×640 inputs of the other configurations and is accounted for when its results are interpreted.

3.5.3 RT-DETR

RT-DETR [Zhao et al. 2023] is the set-prediction representative, evaluated from the `rt detr - l . pt` COCO-pretrained checkpoint. Its efficient hybrid encoder separates intra-scale feature interaction, performed by self-attention on the deepest level only, from cross-scale fusion, performed by convolution, which is what makes transformer-based detection fast enough for this comparison. Because it produces a fixed-size set of predictions without non-maximum suppression, its confidence scores are distributed differently from those of NMS-based detectors, a property that turns out to matter a great deal in Chapter 7.

3.5.4 The Weighted Box Fusion Ensemble

The ensemble comparator combines the predictions of the fine-tuned YOLOv8m and Faster R-CNN models using Weighted Box Fusion [Solovyev, Weimin Wang, and Gabruseva 2021]. Detections from both models are pooled, clustered by overlap, and each cluster is replaced by a single box whose coordinates are the confidence-weighted mean of the cluster members and whose score reflects both the members' scores and the number of models contributing. The relative model weights were swept over $[0.3, 0.7]$. No additional training is involved: the ensemble reuses predictions already produced by its members, so its accuracy is obtained entirely at the cost of running both models.

Table 3.1 summarises all ten configurations evaluated in this dissertation, the six described in this section together with the YOLO-family models of Sections 3.2 and 3.4 and the two single-attention comparators.

3.6 Software Framework and Extension Mechanism

The YOLO-family and RT-DETR configurations were trained and evaluated through the Ultralytics framework [Ultralytics 2023], which builds on PyTorch [Paszke et al.

Configuration	Family	Description
YOLOv8m	One-stage	C2f backbone, PAN-FPN neck, anchor-free decoupled head; approximately 25.9M parameters, COCO-pretrained. Host architecture and principal baseline.
YOLOv10m	One-stage	Consistent dual assignments permitting NMS-free inference; throughput-oriented family member.
YOLOv11m	One-stage	C3k2 blocks with a C2PSA partial self-attention module after SPPF; the host used to test architecture dependence.
Faster R-CNN	Two-stage	ResNet-50-FPN-v2 with region proposal network and RoI feature pooling, COCO-pretrained.
SSD-VGG16	One-stage	Multiscale anchor prediction on a VGG16 backbone at native 300×300 resolution.
RT-DETR	Set prediction	Convolutional backbone, efficient hybrid encoder, transformer decoder; <code>rt detr-1.pt</code> checkpoint, NMS-free.
YOLO+FRCNN (WBF)	Ensemble	YOLOv8m and Faster R-CNN predictions fused by Weighted Box Fusion, model weights swept over $[0.3, 0.7]$.
YOLOv8m+CBAM	Attention	CBAM at the same three neck blocks, same zero-initialised gate and two-phase schedule as the proposed model, without the preceding ECA stage.
YOLOv8m+CoordAttnAttention		Coordinate Attention at the same three neck blocks under the same mechanism and schedule, substituting direction-aware pooling for CBAM's channel-then-spatial refinement.
RGDA-YOLOv8m	Attention	Proposed. Sequential ECA then CBAM at the same three neck blocks, combined with the unrefined features through a learnable scalar gate initialised to zero.

Table 3.1: The ten detector configurations evaluated. All were fine-tuned from COCO-pretrained initialisations on the same 740-image training partition and evaluated on the same 186-image validation partition.

2019] and supplies the C2f blocks, PAN-FPN neck and decoupled head described in Section 3.2 together with the training loop, validation-based checkpoint selection and the `model.val()` evaluation routine. Faster R-CNN and SSD-VGG16 fall outside it and were trained through torchvision with an explicit loop.

Modifying an Ultralytics model conventionally means one of two things, and both were rejected here:

- registering a new block type, which requires editing the framework source;
- writing a modified YAML topology, which shifts layer indices and so silently changes which layers the `freeze` parameter freezes.

Either route changes the model definition, so the modified and unmodified detectors would no longer be guaranteed identical in every respect other than the modification. For an experiment whose purpose is to attribute a sub-percentage-point difference to one module, that is unacceptable.

The mechanism adopted instead is the PyTorch forward hook. Registered on a chosen neck block, it receives that block's output tensor after the forward method returns and may return a replacement, so the attention transformation is applied without the block, the model definition, the layer indexing or the framework source being altered at all, and the two runs differ in precisely one respect. Because the modules are not registered as submodules, their parameters are absent from the checkpoints Ultralytics writes and must be persisted separately. A model distributed as weights alone reverts to an unmodified detector, a limitation revisited in Section 7.8.4.

3.7 Experimental Pipeline and Hardware

Training the ten configurations across multiple seeds involves long-running GPU jobs that can fail part-way through, so the pipeline writes a JSON record per run on completion, holding the configuration, seed, every computed metric, the confidence sweep and the measured throughput, alongside a per-group manifest recording the environment, checkpoint paths and timestamps. An interrupted experiment can therefore resume without repeating completed runs, and every number in Chapter 7 is traceable to the run that produced it. All of those runs used a single NVIDIA GeForce RTX 3090 with 24 GB of memory, and every inference measurement reported in this dissertation uses FP32 precision at batch size one, the operating regime of a real-time streaming application and a deliberately conservative choice, since batching and reduced precision would raise every throughput figure without corresponding to the deployment scenario the work targets. Because all configurations were measured under identical conditions on identical hardware, the relative comparisons hold even though the absolute figures are specific to this GPU.

Chapter 4

Dataset Construction and Analysis

Every number reported in Chapter 7 rests on the corpus described here, and the single most consequential decision in this dissertation was made at this stage rather than at the architectural one. This chapter documents the three source repositories, the deduplication procedure that removed more than half of their union, the unification of two incompatible annotation formats, the partitioning scheme, and the resulting statistical properties of the data. It closes with the threats to validity that the corpus imposes on everything built on it.

4.1 Source Repositories

Three publicly available pothole datasets hosted on Kaggle were consolidated into a single corpus. The first, `chitholian/annotated-potholes-dataset` [Chitholian 2020], contains 665 images with bounding-box annotations in Pascal VOC XML format. The second, `andrewmvd/pothole-detection` [Larxel (AndrewMVD) 2020], also contains 665 images with Pascal VOC XML annotations. The third, `ashishkumarak/training-setzip` [Ashish Kumar 2021], contains 674 images with annotations supplied as CSV bounding-box records.

The union therefore comprises 2,004 annotated images carrying 4,851 ground-truth boxes: 1,740 from each of the two VOC repositories and 1,371 from the CSV repository. Multiple repositories were used deliberately. A single small pothole dataset captures one photographer’s equipment, one geography and one set of lighting conditions, and a detector trained on it learns those incidental regularities alongside the defect itself. Combining independently assembled repositories broadens the appearance distribution, provided the repositories are in fact independent, which is the question the next section addresses.

4.2 Deduplication

4.2.1 Why Byte-Level Hashing Failed

The obvious way to find duplicate images is to hash the files and group the collisions. This was attempted first, using MD5 over the raw file bytes, and it proved unreliable in a specific and instructive way: the same procedure applied to the same three repositories found 674 duplicates in one download session and none in another.

The explanation is that Kaggle packages are not byte-stable. The same photograph redistributed through two repositories is re-encoded (different JPEG quantisation tables, different chroma subsampling, different metadata), so the files differ at the byte level while being visually identical. A hash over the file contents is sensitive to exactly the differences that do not matter and blind to the equivalence that does. Any procedure that depends on byte equality is therefore not merely imperfect here but non-deterministic across download sessions, which is disqualifying for a study that needs to be reproducible.

4.2.2 Normalized-Pixel Nearest-Neighbor Matching

The procedure adopted instead operates on image content rather than file content. Each image is converted to greyscale, resampled to a fixed 64×64 grid using Lanczos interpolation, and flattened to a 4,096-element vector. Two images I_a and I_b are treated as duplicates when the mean absolute difference between their normalized representations falls below a threshold τ on a 0–255 intensity scale:

$$d(I_a, I_b) = \frac{1}{4096} \sum_{k=1}^{4096} |\mathbf{v}_a[k] - \mathbf{v}_b[k]| < \tau, \quad \tau = 1.0. \quad (4.1)$$

Greyscale conversion removes sensitivity to colour-space differences introduced by re-encoding; downsampling to 64×64 removes sensitivity to resolution differences and to the high-frequency artefacts that JPEG compression introduces; and the mean absolute difference is a robust, interpretable measure on a scale that can be reasoned about directly. Images are processed in order, each compared against the set of already-retained representations, and retained only if no retained image lies within τ . Every image appearing in more than one repository is therefore kept exactly once, and its first-encountered copy is the one retained.

4.2.3 Threshold Selection

The threshold $\tau = 1.0$ was set by measuring d on sampled pairs that had been confirmed duplicate by inspection and on sampled unrelated pairs. Confirmed duplicates measured between 0.10 and 0.50 mean absolute intensity difference. Unrelated pairs measured far higher. The separation between the two populations is therefore roughly an order of magnitude, and $\tau = 1.0$ sits comfortably inside the gap rather than at a

Repository	Source	Images	Removed	Retained	Annotation format
chitholian	Kaggle	665	0	665	Pascal VOC XML
andrewmvd	Kaggle	665	656	9	Pascal VOC XML
ashishkumarak	Kaggle	674	422	252	CSV bounding boxes
Total	—	2,004	1,078	926	Unified to YOLO format

Table 4.1: Composition of the corpus before and after normalized-pixel nearest-neighbor deduplication. The union of the three repositories contains 4,851 annotated boxes. The deduplicated corpus contains 2,354.

boundary where small changes in the threshold would change the outcome. This insensitivity matters: a threshold chosen at a point where the result varies rapidly would make the entire corpus, and hence every result in this dissertation, an artefact of an arbitrary constant.

4.2.4 Outcome

The procedure removed 1,078 of the 2,004 images, and the distribution of those removals is the substantive finding: the `andrewmvd` repository proved almost entirely contained within `chitholian`, contributing only 9 unique images against 656 duplicates, a further 422 duplicates were removed from `ashishkumarak` leaving 252, and `chitholian`, processed first, retained all 665 by construction, as Table 4.1 sets out. Had the step been omitted, those 1,078 images, 54 percent of the corpus, would each have had a near-identical twin in the data, the overwhelming majority of which would straddle an 80:20 partition, so the detector would have been evaluated substantially on images it had memorised and every metric would have been inflated invisibly and by a margin far exceeding the sub-percentage-point architectural effects this dissertation sets out to measure. The same lesson was learned once more during the experimental work, when an early set of baselines was inadvertently scored over the entire un-deduplicated corpus and produced a Faster R-CNN recall of 0.9988 against an implied 4,851 ground-truth boxes where the held-out partition contains 474. The pipeline now rejects any baseline whose implied count does not match the partition being scored. Chapter 2 noted that few published pothole-detection studies report a deduplication step, and Section 7.8.3 returns to the general point these two episodes illustrate, that procedural errors in this domain are larger than the architectural ones under study and announce themselves as plausible numbers rather than as failures.

4.3 Annotation Unification and Preprocessing

4.3.1 Coordinate Conversion

The three repositories supply annotations in two formats. Pascal VOC XML records absolute pixel corner coordinates $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$. The CSV repository records bounding boxes as tabular rows. Both were converted to the YOLO convention of normalized centre coordinates and normalized extents $[c_x, c_y, w, h]$ [Ultralytics 2023]:

$$\begin{aligned} c_x &= \frac{x_{\min} + x_{\max}}{2W}, & c_y &= \frac{y_{\min} + y_{\max}}{2H}, \\ w &= \frac{x_{\max} - x_{\min}}{W}, & h &= \frac{y_{\max} - y_{\min}}{H}, \end{aligned} \quad (4.2)$$

where W and H are the original image width and height. Normalization makes the annotation independent of the image resolution, which is necessary because the three repositories do not share one. The corpus contains a single object class, pothole, so no class-taxonomy reconciliation was required, an advantage of pothole-specific data over the multi-class road-damage corpora discussed in Chapter 2, where merging sources requires mapping between incompatible damage taxonomies.

4.3.2 Input Resolution and Augmentation

All models except SSD-VGG16 were trained and evaluated at 640×640 pixels. SSD-VGG16 uses its native 300×300 resolution. Training augmentation used the default Ultralytics pipeline: mosaic composition, which tiles four training images into one and thereby increases the number of objects and the diversity of contexts per training sample; random horizontal flipping; HSV colour jitter, which is the augmentation most relevant to a task confounded by illumination and surface colour; and random scaling. No augmentation was applied at validation. The augmentation policy was held identical across all configurations, so it cannot contribute to differences between them.

4.4 Partitioning

The deduplicated corpus was shuffled and divided into training and validation partitions in an 80:20 ratio using a fixed random seed of 42, giving 740 training images and 186 validation images containing 474 annotated pothole instances. This single canonical split is reused by every configuration and every seed reported in this dissertation. The seed that controls the split is deliberately independent of the seeds that control model initialisation and data ordering: the three-seed protocol of Chapter 6 varies training stochasticity while holding the evaluation set exactly fixed, so that run-to-run differences cannot be attributed to differences in what was being measured.

No separate test partition was held out. With 926 images, splitting three ways would leave a test set too small for the differences under study to be resolvable at all. The

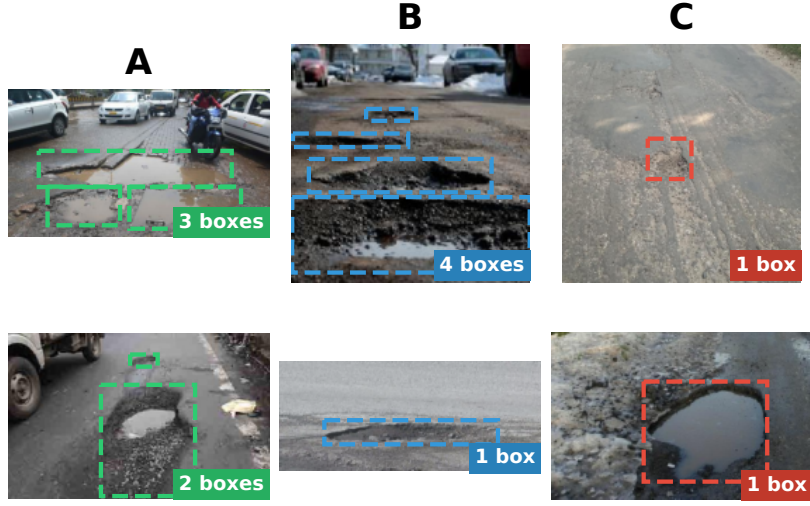


Figure 4.1: Representative annotated images from the three source repositories. Ground-truth annotations are shown in green (chitholian), blue (andrewmvd) and red (ashishkumarak).

consequence is that the validation partition serves both for checkpoint selection in the YOLO-family models and for reporting, which introduces an optimistic bias. That bias applies equally to every configuration scored through `model.val()` and is stated as a limitation in Section 7.8.4.

4.5 Characteristics of the Corpus

The deduplicated corpus contains 926 images and 2,354 annotated pothole instances, an average of 2.54 instances per image, with a maximum of 19 instances in a single image. This distribution is characteristic of the task: most road scenes contain one or two clearly defined defects, while a minority of images of severely deteriorated surfaces contain many small ones. The long tail matters for evaluation, because a handful of dense images contribute a disproportionate share of the total instance count and therefore a disproportionate share of the recall denominator.

Figure 4.1 shows representative annotated images from the three repositories, with ground-truth boxes drawn in green, blue and red respectively for chitholian, andrewmvd and ashishkumarak. The examples illustrate the appearance variation the detector must accommodate: differences in pavement texture and colour, in pothole morphology from shallow surface scaling to deep cavities with undercut edges, in illumination from overcast to strong directional sunlight with cast shadows, and in the number and apparent scale of annotated instances.

Three properties of the corpus bear directly on the architectural design of Chapter 5. First, potholes occupy a small and highly variable fraction of the image area, which is the condition under which multi-scale feature fusion matters and therefore the condition under which a recalibration applied at the fusion stage can be expected to have leverage. Second, the defect boundary is frequently weak: the transition from intact pavement to defect is gradual, textural rather than photometric, and often partly occluded by water or debris, so channel-level discrimination between defect texture and pavement texture is a plausible locus of improvement. Third, the confounders are spatially structured (shadows, patches and wet regions are contiguous areas with definite boundaries, not noise) which is the condition under which spatial attention is a plausible remedy. These three observations are the empirical basis for combining a channel and a spatial mechanism rather than either alone.

4.6 Threats to Validity

Four properties of the corpus limit what can be concluded from it.

Scale. At 926 images and 2,354 instances the corpus is small by the standards of general object detection and small even by the standards of the road-damage benchmarks discussed in Chapter 2. The 186-image validation partition in particular is small enough that individual images influence the reported metrics measurably, which is the principal reason that run-to-run variation is comparable in magnitude to the architectural effects under study.

Condition coverage. All three repositories consist predominantly of daylight, fair-weather photography. Night, heavy rain, snow, fog and low sun angles are essentially absent. Frnda et al. [Frnda et al. 2024] show that detection accuracy degrades sharply and unevenly under exactly these conditions, so no claim made here extends to them.

Viewpoint and geography. The images are predominantly ground-level views from vehicle or pedestrian height. Aerial and UAV perspectives, of the kind represented in UAV-PDD2023 [Pan, Xu Wang, and Huan 2025], are absent, and the geographic distribution of the source repositories is neither documented nor controlled.

Annotation consistency. The three repositories were annotated independently, by different people, without a shared written definition of what constitutes a pothole or where its boundary lies. Deduplication removed the redundancy between repositories but cannot reconcile differences in annotation convention among the images that remain. Some of the residual error of every model evaluated here is therefore annotation noise rather than detection failure, and this contributes equally to all configurations.

Chapter 5

Attention Design and Training Protocol

This chapter presents RGDA-YOLOv8m. It begins with the design rationale (the reasoning that determined where attention is applied, which mechanisms are combined, and why the combination is gated) because those decisions, rather than the modules themselves, constitute the contribution. It then specifies the two attention mechanisms mathematically, defines the combined residual-gated block, accounts for its parameter cost exactly, describes the forward-hook injection mechanism, and sets out the two-phase training protocol and the factorial ablation design.

Figure 5.1 shows the complete architecture and training workflow, and serves as a map for the rest of the chapter. A 640×640 road image enters the C2f-based YOLOv8m backbone, which produces multiscale P3, P4 and P5 features. The PAN-FPN neck fuses these through top-down and bottom-up paths, and the three highlighted neck blocks, with channel dimensions $C \in \{192, 384, 576\}$, are the forward-hook injection points. At each, ECA performs channel recalibration, CBAM applies its channel and spatial gates, and the refined tensor is combined with the original through the zero-initialised residual gate before being passed to the decoupled detection head. The lower part of the figure summarises the two-phase schedule of Section 5.6: the backbone is frozen in Phase 1 and all layers fine-tuned in Phase 2.

5.1 Design Rationale

5.1.1 Where to Apply Attention

An attention module can be inserted almost anywhere in a detector, and the choice is consequential. Three candidate locations were considered.

Applying attention *within the backbone* would recalibrate features during extraction, before any multi-scale fusion. This is where several published road-damage detectors place it [Chen, Yu, and Shu Li 2024; Linxuan Zhang, Deng, and Zou 2025]. The objection

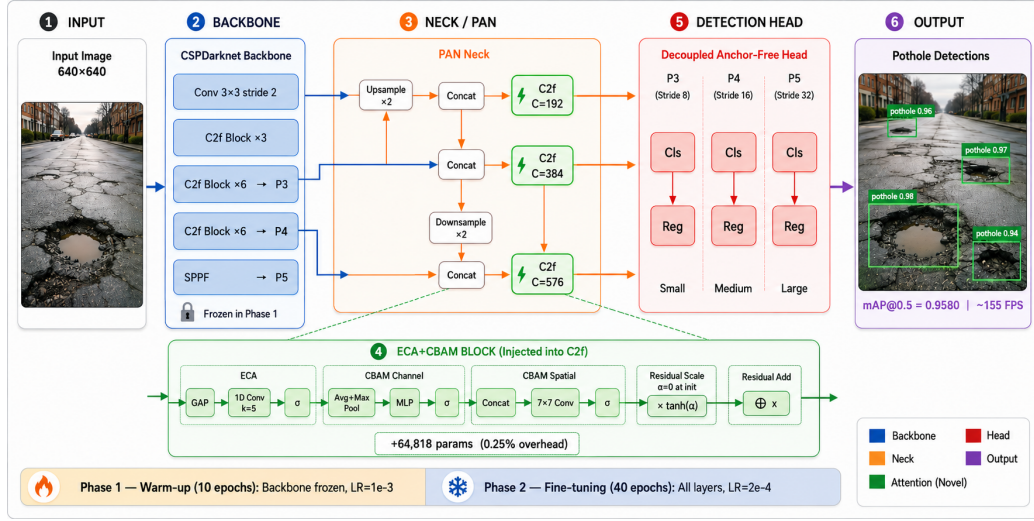


Figure 5.1: Architecture and training workflow of RGDA-YOLOv8m. The three highlighted PAN-FPN neck blocks, with $C \in \{192, 384, 576\}$, are the forward-hook injection points at which the residual-gated ECA–CBAM block is applied.

in this setting is that the backbone carries COCO-pretrained weights that constitute most of the model’s transferable knowledge, and that perturbing its intermediate representations early in fine-tuning on 740 images risks damaging that knowledge faster than the small dataset can rebuild it.

Applying attention *within the detection head* would recalibrate features immediately before prediction. The objection is one of leverage: the head operates on features that are already scale-specialised, so a recalibration there can only reweight what the neck has already decided to pass on.

Applying attention *at the neck outputs*, the position adopted, places it at the point where multi-scale information has been fused but before any prediction has been formed. This is where the representation is richest and where a reweighting therefore has the widest influence: it affects every prediction made at that scale. It is also where the confounders identified in Section 4.5 are most likely to be separable, because the fused representation contains both the fine texture cues that distinguish defect from pavement and the coarse contextual cues that distinguish a defect from a shadow.

Attention is applied at all three neck outputs rather than a subset, and the same configuration is used at each. This is a deliberate constraint: it means the ablation of Section 5.8 varies the *type* of attention while holding its *position* fixed, so any observed difference cannot be attributed to placement.

5.1.2 Which Mechanisms to Combine

Section 4.5 identified two distinct failure modes in this data. Potholes are confused with visually similar road textures (patched asphalt, coarse aggregate, wet surfaces) which is a discrimination problem in the channel dimension, since different channels of a fused feature map respond to different textural and photometric properties. Potholes are also confused with spatially structured confounders (shadows, drainage features, worn lane markings) which is a discrimination problem in the spatial dimension.

A channel mechanism alone can suppress an uninformative feature type but cannot express that the feature is informative in one part of the image and not another. A spatial mechanism alone can emphasise a region but operates on whatever channel mixture it is given, including uninformative channels. The hypothesis motivating this design is that neither is sufficient in isolation for this task and that their composition is: suppressing an uninformative channel helps only if the spatially relevant region is also emphasised, and emphasising a region helps only if the channels describing it are the informative ones. This hypothesis is directly testable, and Section 5.8 sets out the factorial design that tests it.

ECA [Q. Wang et al. 2020] was selected for the channel stage in preference to Squeeze-and-Excitation because it avoids the dimensionality-reduction bottleneck and costs a handful of parameters rather than thousands. CBAM [Woo et al. 2018] was selected for the second stage because it supplies the spatial mechanism. CBAM contains a channel branch of its own, so placing ECA before it is not a redundancy but a specific design choice: the ECA factor in this architecture measures the contribution of an additional, parameter-light, local cross-channel interaction applied *before* CBAM's own bottlenecked channel gate and its spatial refinement. This is stated explicitly because it determines how the ablation must be interpreted.

5.1.3 Why the Pathway Is Gated

A newly attached attention module has randomly initialised weights. On its first forward pass it therefore multiplies a carefully pretrained feature map by an arbitrary pattern of values in $(0, 1)$, and the resulting gradients propagate into the backbone and neck. On a large dataset this is survivable. On 740 images it is a genuine risk: the pretrained representation can be degraded during the first few hundred optimisation steps faster than the available data can restore it, and the resulting model may be worse than the one that was modified.

The remedy adopted is a learnable scalar gate initialised so that the modified network is, at initialisation, exactly the unmodified network. The attention pathway then contributes only to the extent that gradient descent finds it useful to open the gate. This is the same principle that motivates zero-initialised residual scaling in deep network training [Bachlechner et al. 2021] and zero-initialised control branches in conditional generative models [Lvmin Zhang, Rao, and Agrawala 2023]: a new component should begin as an identity and earn its influence.

The gate has a second, methodological virtue. Because the modified model at

initialisation is provably identical to the baseline, any difference in final performance is attributable to what training did with the attention pathway rather than to a different starting point. Without the gate, the modified and unmodified runs would begin from different functions, and the comparison would be correspondingly weaker.

5.1.4 Alternatives Considered and Rejected

Three alternatives were considered. A *fixed* residual weight, such as $\mathbf{out} = \mathbf{x} + \lambda(\text{Att}(\mathbf{x}) - \mathbf{x})$ with λ a hyperparameter, would introduce a constant requiring tuning on a validation set already being used for reporting. A *per-channel* gate would allow the network to admit attention selectively by channel, but adds C parameters per module and, more importantly, makes the initialisation argument harder to state cleanly. A *parallel* rather than sequential arrangement of the channel and spatial mechanisms, in the manner of BAM [Park et al. 2018], was not pursued because the sequential arrangement is what CBAM specifies and changing it would confound the comparison with the CBAM-only ablation condition. The single scalar gate was retained as the minimal device that achieves the required property.

5.2 Efficient Channel Attention

ECA [Q. Wang et al. 2020] produces one multiplicative weight per channel from a global descriptor of that channel and its neighbors. For a feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, global average pooling produces a channel descriptor $\mathbf{z} \in \mathbb{R}^C$:

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W X_c(i, j). \quad (5.1)$$

The descriptor is treated as a one-dimensional signal of length C and convolved with a single learned kernel of size k , after which a sigmoid maps the result to multiplicative weights:

$$\mathbf{a} = \sigma(\text{Conv1D}_k(\mathbf{z})), \quad \hat{\mathbf{X}}_c = a_c \mathbf{X}_c. \quad (5.2)$$

Each channel's weight therefore depends on its own descriptor and those of the $k - 1$ neighboring channels, which is the “local cross-channel interaction” that gives the method its name. The kernel size is set adaptively from the channel dimension, on the reasoning that wider layers require wider interaction:

$$k = \Psi(C) = \left\lceil \left\lfloor \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right\rfloor \right\rceil_{\text{odd}}, \quad \gamma = 2, b = 1, \quad (5.3)$$

where $\lceil \cdot \rceil_{\text{odd}}$ denotes rounding up to the nearest odd integer so that the convolution can be symmetrically padded. For the three YOLOv8m neck injection points, with $C \in \{192, 384, 576\}$, Equation (5.3) yields $k = 5$ in every case. Each ECA module therefore contains five learned parameters, and the three together contain fifteen.

The absence of a bottleneck is the substantive difference from Squeeze-and-Excitation [Hu, Shen, and Sun 2018], whose two fully connected layers with reduction ratio r contain $2C^2/r$ parameters, 4,608 at $C = 192$ against ECA's five, and with every channel interaction forced through a C/r -dimensional bottleneck.

5.3 Convolutional Block Attention Module

CBAM [Woo et al. 2018] refines a feature map in two sequential stages. Writing $\otimes$ for element-wise multiplication with broadcasting:

$$\mathbf{F}' = \mathbf{M}_c(\mathbf{F}) \otimes \mathbf{F}, \quad \mathbf{F}'' = \mathbf{M}_s(\mathbf{F}') \otimes \mathbf{F}'. \quad (5.4)$$

The channel gate $\mathbf{M}_c \in \mathbb{R}^{C \times 1 \times 1}$ extends Squeeze-and-Excitation by summing the responses of a shared two-layer perceptron to both average-pooled and max-pooled descriptors, on the reasoning that average pooling captures the extent of a response while max pooling captures its peak:

$$\mathbf{M}_c(\mathbf{F}) = \sigma(\text{MLP}(\text{AvgPool}(\mathbf{F})) + \text{MLP}(\text{MaxPool}(\mathbf{F}))). \quad (5.5)$$

The perceptron is implemented as two 1×1 convolutions with a ReLU between them, reducing to C/r channels and restoring to C , with reduction ratio $r = 16$ throughout this work.

The spatial gate $\mathbf{M}_s \in \mathbb{R}^{1 \times H \times W}$ pools across the channel dimension rather than the spatial one, producing two single-channel maps that are concatenated and convolved:

$$\mathbf{M}_s(\mathbf{F}) = \sigma(f^{7 \times 7}[\text{AvgPool}_c(\mathbf{F}); \text{MaxPool}_c(\mathbf{F})]). \quad (5.6)$$

The 7×7 kernel is deliberately large: the spatial gate must decide whether a region is a defect or a confounder, and that decision requires context beyond the immediate neighborhood.

5.4 The Residual-Gated ECA–CBAM Block

5.4.1 Composition

The proposed block applies ECA, then CBAM's channel gate, then CBAM's spatial gate, and combines the result with the untransformed input through a learnable scalar gate. Writing $\mathbf{A}(\mathbf{x})$ for the composed attention transformation,

$$\mathbf{A}(\mathbf{x}) = \mathbf{M}_s(\mathbf{u}) \otimes \mathbf{u}, \quad \mathbf{u} = \mathbf{M}_c(\mathbf{e}) \otimes \mathbf{e}, \quad \mathbf{e} = \text{ECA}(\mathbf{x}), \quad (5.7)$$

the block output is

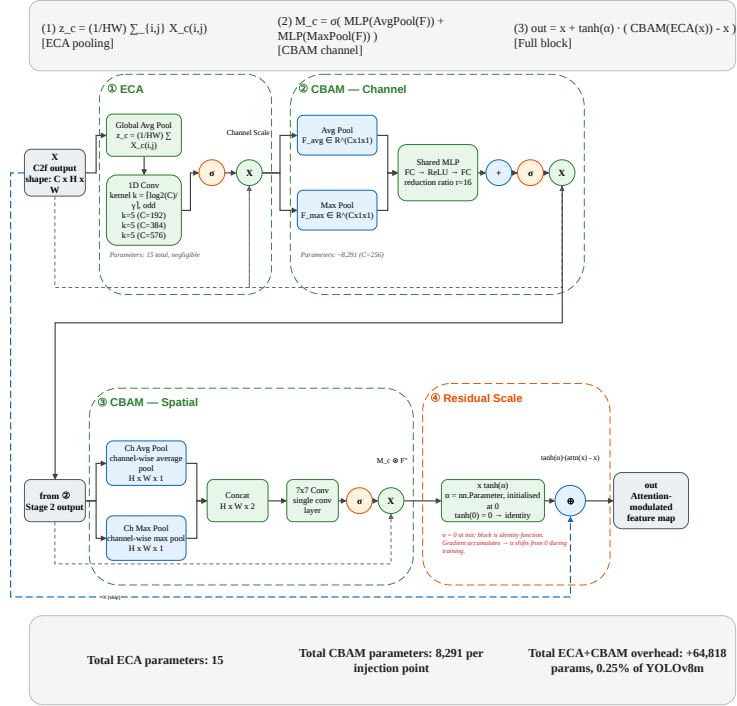


Figure 5.2: The residual-gated ECA-CBAM block. The input passes through ECA channel recalibration, then CBAM’s channel and spatial gates. The refined output is combined with the unmodified input through the learnable gate α , initialised to zero.

$$\text{out} = \mathbf{x} + \tanh(\alpha) [\mathbf{A}(\mathbf{x}) - \mathbf{x}], \quad (5.8)$$

where $\alpha \in \mathbb{R}$ is a single learnable parameter initialised to zero.

Figure 5.2 shows the internal structure.

5.4.2 Properties of the Gate

Equation (5.8) is written as an interpolation rather than as a plain residual sum for a specific reason. Because $\tanh(0) = 0$, at initialisation

$$\text{out} = \mathbf{x} + 0 \cdot [\mathbf{A}(\mathbf{x}) - \mathbf{x}] = \mathbf{x}, \quad (5.9)$$

so the modified detector computes exactly the same function as the pretrained one. Had the block been written as $\mathbf{x} + \tanh(\alpha)\mathbf{A}(\mathbf{x})$, zero initialisation would give $\text{out} = \mathbf{x}$ as well, but the gate would then interpolate between the identity and the identity-plus-attention rather than between the identity and attention itself, and $\tanh(\alpha) = 1$ would double the feature magnitude. The interpolation form makes $\tanh(\alpha) = 1$ correspond to

Injection point	C	k	ECA	Channel gate	Spatial gate	Total
Shallow (P3 fusion)	192	5	5	4,608	98	4,712
Middle (P4 fusion)	384	5	5	18,432	98	18,536
Deep (P5 fusion)	576	5	5	41,472	98	41,576
Total	—	—	15	64,512	294	64,824

Table 5.1: Parameter cost of the residual-gated ECA-CBAM block at each of the three neck injection points. Each row includes one parameter for the gate α . The total of 64,824 parameters represents a 0.25 percent increase over the 25.9-million-parameter YOLOv8m base model.

full attention and $\tanh(\alpha) = 0$ to none, which is the intended semantics.

The $\tanh$ envelope bounds the gate to $(-1, 1)$, which prevents the optimiser from driving the attention contribution to an arbitrarily large magnitude and destabilising training, a real possibility given that α is a single unregularised scalar receiving gradient from every spatial position of every channel. The negative half of the range is available and meaningful: it allows the network to apply the attention correction in the opposite direction if that is what reduces the loss.

The gradient with respect to α is

$$\frac{\partial \mathcal{L}}{\partial \alpha} = (1 - \tanh^2(\alpha)) \left\langle \frac{\partial \mathcal{L}}{\partial \mathbf{out}}, A(\mathbf{x}) - \mathbf{x} \right\rangle, \quad (5.10)$$

which is maximal at $\alpha = 0$ and vanishes as $|\alpha|$ grows. The gate therefore moves fastest when it is closed, which is the desirable behaviour: it opens readily if the attention pathway is useful and saturates rather than diverging if it is.

5.4.3 Parameter Accounting

The block’s parameter cost is small and exactly computable. Table 5.1 gives the breakdown at each injection point. Writing $m = \lfloor C/r \rfloor$ for the CBAM bottleneck width with $r = 16$, each block contains k parameters for the ECA convolution, $2Cm$ for the two bias-free 1×1 convolutions of the channel gate, $2 \times 7 \times 7 = 98$ for the bias-free spatial convolution, and one for α .

Two features of this accounting are worth noting. The channel gate accounts for 99.5 percent of the added parameters, because its cost grows as C^2/r while the other components are constant or logarithmic in C . And the deep injection point alone accounts for 64 percent of the total, for the same reason. The entire mechanism costs 0.25 percent of the base model, which places any observed improvement in a different category from one obtained by scaling the model up.

5.5 Hook-Based Neck Injection

5.5.1 Layer Selection

The injector locates candidate blocks by scanning the detector’s top-level module sequence for blocks whose class name belongs to $\{C3k2, C2f, C2fAttn, RepC3, C3\}$. This set spans the fusion block types used across the YOLO versions under study, which is what allows the identical injector to be applied to both YOLOv8m and YOLOv11m in Section 7.4 without modification, a necessary condition for that comparison to be controlled.

Blocks of these types occur in both the backbone and the neck. The selection rule retains the *last three* matches. This is correct because the Ultralytics model sequence is ordered from input to output: backbone blocks necessarily precede neck blocks, and the three PAN-FPN fusion blocks that feed the detection head are the final three matches in the sequence. The rule is positional rather than name-based, which makes it robust to the naming differences between YOLO versions.

5.5.2 Channel Probing

The number of output channels of each selected block is determined empirically rather than assumed. Temporary probe hooks are registered on the three candidate blocks, a dummy tensor is passed through the model, each probe records the channel dimension of the tensor its block emitted, and the probes are removed. The recorded dimensions (192, 384 and 576 for YOLOv8m) are then used to construct correctly sized attention modules.

This step exists because block output width depends on the model scale multipliers, which differ between the n, s, m, l and x variants of each YOLO version. Hard-coding the widths would silently produce a shape mismatch, or worse a silently incorrect module, on any variant other than the one the constants were written for. Probing makes the injector correct by construction across scales and versions.

5.5.3 The Hook

A forward hook is then registered on each selected block. When the block returns, the hook receives its output tensor, applies the residual-gated transformation of Equation (5.8), and returns the result, which PyTorch substitutes for the original output before it reaches the next module. Where a block returns a tuple rather than a tensor, the hook transforms the first element and reconstructs the tuple.

5.5.4 Consequences of the Mechanism

The advantage of this mechanism was stated in Section 3.6: the model definition, layer indices and framework source are untouched, so the modified and unmodified

configurations are identical in every respect except the transformation applied at three points.

One further implementation detail is necessary for the mechanism to work at all. Registering the hooks on the model object before invoking Ultralytics training has no effect, because `YOLO.train()` constructs its own `nn.Module` for the trainer and only reassigns it onto the model object after training completes. The hooked object is therefore discarded the moment training begins. The failure is silent, every configuration trains and evaluates successfully and every configuration reproduces the unmodified baseline, which is easily misread as the attention mechanism having no effect. The hooks are consequently attached from an `on_train_start` callback, which fires after the live trainer model has been built. This is documented here because the same trap would affect any attempt to reproduce or extend this work.

The cost is that the attention parameters live outside the detector's module tree and therefore outside its `state_dict`. They must be persisted separately by the training pipeline and reattached explicitly before inference. A checkpoint written by Ultralytics and loaded without the accompanying attention state produces an unmodified YOLOv8m that runs without error and reports plausible results, which is a failure mode that is silent rather than loud. This is the principal implementation limitation of the work and is revisited in Section 7.8.4.

5.6 Two-Phase Training Protocol

5.6.1 Motivation

Fine-tuning a COCO-pretrained detector on 740 domain-specific images with newly attached, randomly initialised modules poses a scheduling problem. Training everything at once at a learning rate high enough to adapt the new modules risks disturbing the pretrained backbone. Training at a rate low enough to protect the backbone leaves the new modules undertrained. The protocol adopted separates these concerns in time.

5.6.2 Phase 1: Backbone Stabilisation

The first ten layers of the detector are frozen, and the attention modules together with the remaining layers and the detection head are trained for ten epochs at an initial learning rate of 1×10^{-3} with weight decay 5×10^{-4} . Freezing the early backbone protects the general-purpose low-level features that transfer well from COCO, while allowing the head to adapt to a single-class problem with a very different object-scale distribution from the pretraining data.

5.6.3 Phase 2: Full-Model Fine-Tuning

The best checkpoint from Phase 1 is loaded into a fresh YOLO object, a newly initialised attention module is attached at the same three locations, all layers are unfrozen, and

Parameter	Phase 1	Phase 2
Epochs	10	40
Initial learning rate	1×10^{-3}	2×10^{-4}
Batch size	16	16
Input resolution	640×640	640×640
Frozen layers	10	0
Data-loader workers	2	2
Weight decay	5×10^{-4}	5×10^{-4}
Attention module	ϕ_1 (discarded)	ϕ_2 (retained)

Table 5.2: Two-phase training configuration for RGDA-YOLOv8m. The same schedule was applied to the YOLOv8m+CBAM and YOLOv8m+CoordAttn comparators and to every cell of the ablation, so that differences between them reflect the attention configuration rather than the optimisation settings.

training continues for forty epochs at a reduced initial learning rate of 2×10^{-4} . The reduction by a factor of five reflects that the model now begins from a domain-adapted rather than a COCO-pretrained state, and that the backbone is no longer protected by freezing.

Table 5.2 summarises both phases.

5.6.4 Why the Phase-1 Attention Module Is Discarded

A fresh YOLO object is instantiated for Phase 2 to avoid trainer-state conflicts that arise when Ultralytics training is invoked twice on the same object. Because the attention parameters are external to the checkpoint, as Section 5.5 explained, they do not survive that reload: the Phase-1 module ϕ_1 , including its learned gate α , is discarded, and a new module ϕ_2 with $\alpha = 0$ is attached for Phase 2.

The consequence should be stated plainly rather than glossed. Phase 1 does not pre-train the attention pathway. What it produces is a set of backbone and head parameters that have been adapted to the pothole domain *under an attention-modulated training loss*, and Phase 2 trains a new attention module jointly with that adapted detector from a closed gate. The protocol is therefore best understood as domain adaptation followed by gated attention training, not as a two-stage curriculum for the attention module. Whether carrying ϕ_1 into Phase 2 would perform better is an open question that this design does not answer. It is listed as future work in Section 8.5.

5.6.5 Comparator Training

The YOLOv8m+CBAM and YOLOv8m+CoordAttn comparators, and every cell of the ablation, were trained under exactly this schedule from the same COCO-pretrained YOLOv8m initialisation. YOLOv8m, YOLOv10m and YOLOv11m baselines were fine-

tuned for 50 epochs at 640×640 with batch size 16 under Ultralytics automatic optimiser selection with $\text{lr} = 0.01$. RT-DETR was fine-tuned for 50 epochs at batch size 8 from the `rt detr - l . pt` checkpoint, with automatic selection choosing AdamW at learning rate 0.002 and momentum 0.9. Faster R-CNN and SSD-VGG16 were fine-tuned for 20 epochs with SGD at momentum 0.9 and weight decay 0.0005. Faster R-CNN used learning rate 0.005 at batch size 4, while SSD-VGG16 required learning rate 0.0005, batch size 1 and gradient-norm clipping at 1.0 to remain numerically stable. The latter two were trained without validation-based checkpoint selection, so their final-epoch weights were retained. These budget differences are a genuine limitation of the cross-architecture comparison and are treated as such in Section 7.8.4.

5.7 Algorithmic Summary

Algorithm 1 states the forward pass of RGDA-YOLOv8m, including the residual-gated operation applied at each injection point. Algorithm 2 states the multi-seed two-phase training procedure.

5.8 Ablation Design

The hypothesis of Section 5.1, that neither the channel nor the spatial mechanism is sufficient alone and that the effect belongs to their combination, is tested by a 2×2 factorial design over the presence of ECA and the presence of CBAM, giving the four conditions of Table 5.3.

	Without CBAM	With CBAM
Without ECA	Baseline (no attention)	CBAM-only
With ECA	ECA-only	ECA+CBAM (proposed)

Table 5.3: The 2×2 factorial ablation over the presence of the channel and channel–spatial pathways. All four conditions share the dataset partition, injection points, gate, training schedule, confidence threshold and IoU threshold.

All four conditions use the same partition, the same three injection points, the same zero-initialised gate, the same two-phase schedule, the same confidence threshold of 0.25 and the same IoU threshold of 0.5. The only difference is which transformation the gate interpolates towards. Each condition is run at all three seeds and reported as a mean with standard deviation.

Because CBAM contains a channel branch of its own (Equation (5.5)), the ECA factor does not measure the contribution of channel attention in general. It measures the contribution of an *additional*, parameter-light, local cross-channel interaction applied before CBAM’s bottlenecked channel gate. A reader interpreting the ablation as “channel

Algorithm 1 Forward pass of RGDA-YOLOv8m**Input:** Image $I \in \mathbb{R}^{H \times W \times 3}$; detector parameters θ ; attention parameters ϕ **Output:** Detections $\{(\text{box}_i, \text{class}_i, \text{conf}_i)\}$ **Preprocessing**1: $I' \leftarrow \text{Normalize}(\text{Resize}(I, 640 \times 640))$ **Backbone and neck**2: $\{P_3, P_4, P_5\} \leftarrow \text{Backbone}_\theta(I')$ $\triangleright$ C2f multiscale features3: $\{N_3, N_4, N_5\} \leftarrow \text{Neck}_\theta(P_3, P_4, P_5)$ $\triangleright$ PAN-FPN fusion**Residual-gated attention at each neck output**4: **for** each scale $s \in \{3, 4, 5\}$ **do**5: $X \leftarrow N_s$ 6: $\mathbf{z} \leftarrow \text{GlobalAvgPool}(X)$ 7: $\omega \leftarrow \sigma(\text{Conv1D}_k(\mathbf{z}))$ 8: $X_1 \leftarrow X \otimes \omega$ $\triangleright$ ECA channel recalibration9: $\mathbf{M}_c \leftarrow \sigma(\text{MLP}(\text{AvgPool}(X_1)) + \text{MLP}(\text{MaxPool}(X_1)))$ 10: $X_2 \leftarrow X_1 \otimes \mathbf{M}_c$ $\triangleright$ CBAM channel gate11: $\mathbf{M}_s \leftarrow \sigma(f^{7 \times 7}[\text{AvgPool}_c(X_2); \text{MaxPool}_c(X_2)])$ 12: $X_3 \leftarrow X_2 \otimes \mathbf{M}_s$ $\triangleright$ CBAM spatial gate13: $N_s \leftarrow X + \tanh(\alpha_s)(X_3 - X)$ $\triangleright$ gate; $\alpha_s = 0$ at initialisation14: **end for****Detection and post-processing**15: $R \leftarrow \text{DetectionHead}_\theta(N_3, N_4, N_5)$ 16: $R \leftarrow \text{ApplyConfidenceThreshold}(R, \tau_{\text{conf}})$ 17: **return** NonMaxSuppression($R, 0.45$)

attention versus spatial attention” would draw the wrong conclusion from it, which is why the point is made here as well as in Chapter 7.

Algorithm 2 Two-phase multi-seed training of RGDA-YOLOv8m

Input: Deduplicated corpus D ; COCO-pretrained θ_0 ; seeds $S = \{42, 123, 456\}$ **Output:** Trained parameters and aggregated metrics

- 1: $(D_{\text{train}}, D_{\text{val}}) \leftarrow \text{Split}(D, 0.8, \text{seed} = 42)$ $\triangleright$ fixed across all seeds
 - 2: **for** each seed $s \in S$ **do**
 - 3: SetRandomState(s); $\theta \leftarrow \theta_0$
 - 4: ***Phase 1: backbone stabilisation***
 - 5: $\phi_1 \leftarrow \text{AttachAttention}(\theta, \{N_3, N_4, N_5\})$; freeze the first 10 layers $\triangleright \alpha = 0$
 - 6: Train $(\theta_{\text{unfrozen}}, \phi_1)$ for 10 epochs at $\text{lr}_0 = 1 \times 10^{-3}$
 - 7: $\theta_1 \leftarrow$ best-validation checkpoint; discard ϕ_1
 - 8: ***Phase 2: full-model fine-tuning***
 - 9: Instantiate a fresh detector from θ_1 ; $\phi_2 \leftarrow \text{AttachAttention}(\theta_1, \{N_3, N_4, N_5\})$ $\triangleright \alpha = 0$ again
 - 10: Unfreeze all layers and train (θ, ϕ_2) for 40 epochs at $\text{lr}_0 = 2 \times 10^{-4}$
 - 11: $M_s \leftarrow \text{Evaluate}(\theta, \phi_2, D_{\text{val}})$; persist θ, ϕ_2 and M_s to disk
 - 12: **end for**
 - 13: **return** $\text{mean}(M_s), \text{std}(M_s)$ over $s \in S$
-

Chapter 6

Evaluation Metrics and Measurement Protocol

Chapter 2 argued that published road-defect detection results are frequently incomparable because the procedures that produced them are unstated. This chapter specifies, for this dissertation, every choice that affects a reported number: how predictions are matched to ground truth, how precision, recall, F1 and average precision are computed, at which operating points they are reported, how throughput is measured, and how variation across training runs is characterised. The purpose is to make the results of Chapter 7 reproducible and to make explicit which comparisons they support.

6.1 Matching Predictions to Ground Truth

Every detection metric begins with a decision about which predicted boxes count as correct. Overlap is measured by intersection over union: for a predicted box B_p and a ground-truth box B_g ,

$$\text{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}, \quad (6.1)$$

which is one when the boxes coincide, zero when they are disjoint, and scale-invariant. A prediction is eligible to be a true positive when its IoU with some ground-truth box is at least 0.5.

Eligibility is not sufficient, because several predictions may overlap the same annotation. The matching rule used throughout is the greedy one adopted by the PASCAL VOC and COCO reference implementations [Everingham et al. 2010; Lin, Maire, et al. 2014]. Detections are sorted by descending confidence. Each detection in turn is assigned to the highest-IoU ground-truth box that is still unmatched and that meets the threshold. That box is then removed from consideration. A detection for which no unmatched ground-truth box meets the threshold is a false positive. A ground-truth box that remains unmatched when every detection has been processed is a false negative.

Two properties of this rule matter for interpretation. Confidence ordering means a lower-scoring detection cannot displace a higher-scoring one from an annotation, so duplicate detections of the same pothole are counted as false positives, which is the intended behaviour, since a system that reports one defect three times is reporting incorrectly. And because each annotation can be claimed only once, the number of true positives is bounded above by the number of annotations, so recall is well defined. This identical rule was applied to every configuration in this study, including those scored outside the Ultralytics evaluator.

6.2 Precision, Recall and F1

Given counts of true positives TP , false positives FP and false negatives FN :

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP + FP}, & \text{Recall} &= \frac{TP}{TP + FN}, \\ \text{F1} &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \end{aligned} \quad (6.2)$$

For this application the two error types have different practical costs, and neither dominates. A false negative is an unreported hazard: a pothole that a maintenance survey misses, or a defect an assistive system fails to warn about. A false positive is a wasted inspection or, in the assistive case, a spurious alert. A system that alerts on every shadow will be switched off, at which point its recall becomes irrelevant. F1, as their harmonic mean, penalises a detector that achieves a high value of one by sacrificing the other, and is therefore the single-number summary reported here alongside mAP.

6.3 Average Precision

6.3.1 The Precision–Recall Curve

Precision and recall as defined above describe a detector at one confidence threshold. Sweeping the threshold from high to low traces a precision–recall curve: at a high threshold few detections are emitted, so precision is typically high and recall low. As the threshold falls, recall rises and precision generally falls. Average precision summarises the whole curve and is therefore threshold-independent, which is why it is the primary metric for comparing detectors.

6.3.2 Interpolation and Integration

The raw curve is not monotone, admitting one more detection can raise precision if it is correct, so it is first replaced by its monotonically non-increasing envelope:

$$p_{\text{env}}(r) = \max_{r' \geq r} p(r'). \quad (6.3)$$

Average precision is then the area under that envelope:

$$\text{AP} = \int_0^1 p_{\text{env}}(r) \, dr, \quad (6.4)$$

and mean average precision is the mean of AP_c over classes $c \in \mathcal{C}$. This corpus contains a single class, so mAP@0.5 equals AP at an IoU threshold of 0.5. The notation is retained for consistency with the literature. Where reported, mAP@0.5:0.95 is the mean of AP computed at IoU thresholds from 0.50 to 0.95 in steps of 0.05, which rewards precise localisation rather than mere detection.

6.3.3 Which Interpolation Rule

Equation (6.4) is evaluated by trapezoidal integration over 101 uniformly spaced recall values in $[0, 1]$, which is the COCO convention [Lin, Maire, et al. 2014] and the definition implemented by `Ultralytics model.val()`. It is not the only convention in use. The PASCAL VOC 2007 rule averages interpolated precision at eleven recall levels $\{0, 0.1, \dots, 1.0\}$ [Everingham et al. 2010], and yields systematically different values from the same predictions.

The difference is not negligible. Scoring the same YOLOv11m+ECA+CBAM predictions under both rules gives a mean mAP@0.5 of 0.7555 under the VOC 11-point rule and 0.7823 under COCO 101-point interpolation, a gap of 2.7 percentage points, more than three times the size of the architectural effect this dissertation sets out to measure. A paper reporting “ mAP@0.5 ” without stating the rule has therefore left its reader unable to compare its number with anyone else’s. The COCO rule is used exclusively here, and applied identically to every configuration.

That last point requires enforcement rather than assertion. Four configurations (Faster R-CNN, SSD-VGG16, RT-DETR and the WBF ensemble) are not scored by `model.val()` but by a custom evaluator, because they are either not Ultralytics models or, in the ensemble’s case, not a model at all. That evaluator implements Equations (6.1) to (6.4) directly, so its outputs are commensurable with those of `model.val()`.

Two boundary conventions complete the specification. Precision–recall curves are generated from detections produced at a confidence threshold of 0.001, so that integration covers the full curve rather than a truncated segment of it. A higher threshold would silently discard the low-precision, high-recall tail and inflate AP. And a configuration for which no detection survives the threshold is assigned $\text{AP} = 0$ rather than having Equation (6.4) evaluated from boundary conditions, which would otherwise produce an artificially non-zero value from an empty curve.

6.3.4 Why mAP@0.5 Is the Primary Metric

Two IoU regimes are reported in this dissertation, and their roles differ. mAP@0.5 asks whether the detector found the defect. MAP@0.5:0.95 asks additionally how precisely it delineated it. For general object detection the stricter metric is usually preferred, because a tight box on a rigid object is both achievable and useful.

For potholes it is less informative, for the reason given in Section 1.1.1: the defect boundary is genuinely gradual, so where the annotator drew the box is partly a judgement, and the three source repositories were annotated independently without a shared convention. A metric that rewards agreement with the annotator's boundary at IoU 0.9 is therefore partly measuring annotation style. mAP@0.5 is reported as the primary metric because the downstream tasks (warning a pedestrian, flagging a location for inspection) require knowing that a defect is present and roughly where, not its exact extent. mAP@0.5:0.95 is reported alongside it in Table 7.2 so that a reader interested in localisation quality can see it, and because a change that improves the loose metric while degrading the strict one would indicate that the model had learned to emit larger boxes rather than better ones.

6.4 Operating Points

6.4.1 The Problem With a Single Threshold

Precision, recall and F1 require a confidence threshold. Comparing detectors at a common fixed threshold appears fair and often is not, because detectors differ in how they distribute confidence. A model whose scores are compressed towards the top of the range and one whose scores are spread across it will occupy very different positions on their respective precision–recall curves at the same numerical threshold, and the resulting F1 comparison will conflate detection capability with score calibration.

This is not hypothetical in the present study. The F1-maximising threshold varies between 0.10 and 0.70 across the ten configurations evaluated. A single fixed threshold therefore places some detectors near their optimum and others far from it.

6.4.2 Two Reported Operating Points

Two operating points are consequently reported for every configuration.

The first is a **fixed threshold of 0.25**, applied uniformly. This represents a deployment setting in which one threshold is chosen and applied without per-model tuning, and it is the operating point at which precision, recall and F1 are reported unless stated otherwise.

The second is **F1***, the maximum F1 attained over a sweep of the confidence threshold across [0.05, 0.70]. This characterises the detector at the best point on its own curve and is therefore independent of calibration.

Reporting both separates two questions that a single number conflates: how well does this detector detect, and how well does its confidence scale align with a conventional threshold. Chapter 7 contains a case, RT-DETR, where the two answers differ by 11.1 percentage points of F1, which is larger than the entire spread of the remaining configurations.

6.5 Throughput Measurement

Frames per second is the most frequently misreported quantity in this literature, because it depends on a set of choices that are rarely stated. Each is specified below.

6.5.1 Definition

Throughput is measured over a dedicated pass of 100 validation images and computed as total images divided by total elapsed wall-clock time:

$$\text{FPS} = \frac{N}{\sum_{i=1}^N t_i}, \quad (6.5)$$

and *not* as the mean of per-image reciprocal latencies,

$$\widetilde{\text{FPS}} = \frac{1}{N} \sum_{i=1}^N \frac{1}{t_i}. \quad (6.6)$$

Because the reciprocal is convex, Jensen's inequality gives $\widetilde{\text{FPS}} \geq \text{FPS}$, with equality only if every t_i is identical. The second definition is therefore biased upward, and the size of the bias grows with the variance of the per-image times, so it flatters exactly those models whose latency is least predictable, which is the opposite of what a real-time evaluation should reward. Equation (6.5) answers the operationally meaningful question: how many frames can this detector process per second of sustained operation.

6.5.2 Warm-Up and Synchronisation

The first five iterations of every measurement are discarded, leaving 95 timed forward passes per configuration. Initial forward passes include CUDA context creation, kernel autotuning, memory-pool allocation and lazy weight transfer, none of which recur in steady-state operation.

CUDA operations are asynchronous with respect to the host: a Python statement that launches a kernel returns before the kernel completes. Reading the clock without synchronising therefore measures launch time rather than execution time, and produces throughput figures that can be wrong by an order of magnitude. Explicit device synchronisation is performed after every forward pass so that the timer captures completed work.

6.5.3 Scope of the Timed Region

Timing covers the forward pass and non-maximum suppression. Image decoding is performed before the timer starts, because decoding is a CPU-bound cost determined by file format and storage medium rather than by the detector, and including it measures the test harness rather than the model.

Non-maximum suppression is included because it is a mandatory part of producing detections for every configuration that uses it, and because its cost is not constant: it scales with the number of candidate boxes surviving the confidence threshold. For the YOLO-family and RT-DETR configurations, throughput was measured at the same confidence threshold of 0.001 used to generate the precision–recall curves. This is deliberately conservative, it maximises the number of surviving boxes and therefore the suppression cost, and it means the reported figures are lower bounds relative to a deployment threshold of 0.25. Faster R-CNN and SSD-VGG16 apply an internal score threshold that is not exposed as an inference parameter, so this control could not be applied to them.

6.5.4 Measurement Conditions

All measurements used the hardware and precision described in Section 3.7. Because all configurations were measured under identical conditions in the same session, relative comparisons are valid even though absolute figures are hardware-specific.

Section 7.8.3 quantifies what happens when these choices are made differently: measuring inside an evaluation loop, with decoding interleaved, understated throughput by 20 to 26 percent and inverted the ordering of the YOLO variants relative to one another.

6.6 Multi-Seed Protocol and Reporting

Deep detectors fine-tuned on small datasets vary between runs. The sources are the random initialisation of any newly added parameters, the shuffling of the training data, the sampling of stochastic augmentations such as mosaic composition, and non-deterministic reduction order in GPU kernels. On 740 training images this variation is not negligible relative to the effects under study.

The proposed model and every cell of the ablation were therefore trained at three seeds, $S = \{42, 123, 456\}$, and are reported as mean $\pm$ sample standard deviation. The seed controlling the train/validation split is fixed independently at 42 for every run, so the evaluation set is identical throughout and seed-to-seed differences reflect training stochasticity alone.

Three seeds is a small sample and its limitations are stated rather than elided. It supports a claim that an effect is consistent in *sign* across runs, a paired comparison in which the modified configuration beats its baseline at every seed is meaningful evidence, but it does not support a claim about the *magnitude* of the effect, and it

does not license significance testing. Where a mean difference is smaller than the corresponding standard deviation, Chapter 7 says so explicitly rather than presenting the difference as established.

The cross-architecture comparators of Table 3.1 are single training runs conducted as separate experiments, for reasons of computational budget. This is a real asymmetry: differences between nominally identical backbones trained in different experiments are of the same order as the effects being compared. It is the reason the controlled comparison for the attention mechanism is the ablation of Section 7.3, in which every condition is trained under one schedule at three seeds, rather than the cross-architecture table.

Chapter 7

Results and Discussion

This chapter reports and interprets the experimental results. It proceeds in six stages. Section 7.1 compares RGDA-YOLOv8m with the nine comparator configurations on the common validation partition. Section 7.2 characterises the proposed model’s variability across three random seeds. Section 7.3 isolates the individual and joint contributions of the two attention pathways. Section 7.4 applies the identical block to YOLOv11m to test whether the effect is a property of the module or of the pairing. Section 7.5 examines the learned spatial attention qualitatively, and Section 7.6 analyses the resulting accuracy–throughput trade-off. Section 7.8 then discusses what these results collectively establish, what they do not, and what limits them.

All results use the 186-image validation partition described in Chapter 4, an IoU threshold of 0.5, and the evaluation protocol specified in Chapter 6. Precision, recall and F1 are reported at a fixed confidence threshold of 0.25 unless stated otherwise.

7.1 Cross-Architecture Benchmark

7.1.1 Overall Comparison

Table 7.1 reports the ten configurations, and Figure 7.1 presents the same data on the three headline measures.

No single configuration leads on all three measures. The WBF ensemble leads mAP@0.5, RT-DETR leads F1*, and RGDA-YOLOv8m leads F1 at the fixed operating point. This dispersion is itself a result: it indicates that the ranking of detectors on this task depends on which question is being asked, and that a paper reporting a single metric can present almost any of these architectures as the leader.

7.1.2 The Proposed Model

RGDA-YOLOv8m attains the highest F1 in the table at 0.7708, with an mAP@0.5 of 0.8141 at 139.9 FPS. Among configurations exceeding 100 FPS it has the highest mAP@0.5,

Configuration	mAP@0.5	Precision	Recall	F1	F1*	FPS
YOLO+FRCNN (WBF)	0.8272	0.8564	0.6793	0.7576	0.7774	23.6
RT-DETR	0.8174	0.5674	0.8523	0.6813	0.7919	60.5
YOLOv8m	0.8100	0.7783	0.7553	0.7666	0.7755	152.2
Faster R-CNN	0.7976	0.7389	0.7342	0.7365	0.7399	28.1
YOLOv11m	0.7899	0.7419	0.7278	0.7348	0.7446	160.6
YOLOv8m+CoordAttn	0.7879	0.7543	0.7384	0.7463	0.7495	147.3
YOLOv8m+CBAM	0.7782	0.7400	0.7447	0.7424	0.7541	142.0
YOLOv10m	0.7598	0.7014	0.7236	0.7124	0.7235	172.8
SSD-VGG16	0.6165	0.6936	0.5253	0.5978	0.6169	174.7
RGDA-YOLOv8m (proposed)	0.8141	0.7922	0.7510	0.7708	0.7711	139.9

Table 7.1: Performance of the ten configurations on the 186-image validation partition. Precision, recall and F1 are at a fixed confidence threshold of 0.25. F1* is the maximum F1 over a confidence sweep. Bold marks the best value in each column. The proposed model's row is a mean over three seeds. All comparators are single runs, so small differences between nominally identical backbones reflect run-to-run variation.

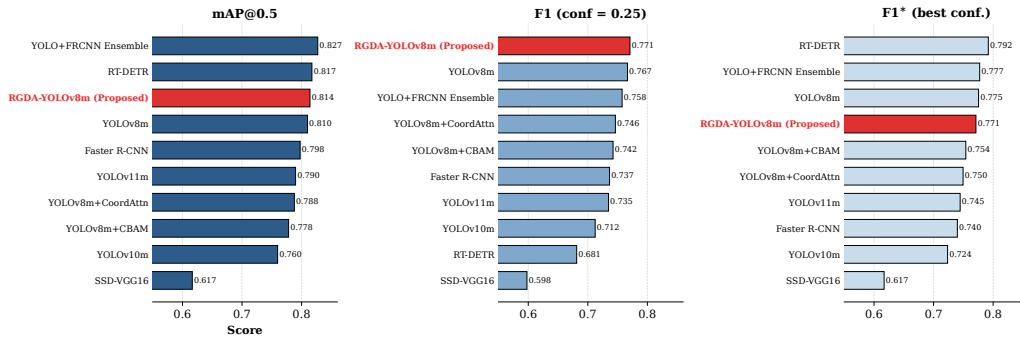


Figure 7.1: mAP@0.5, F1 at a fixed confidence threshold of 0.25, and F1* at each model's best confidence, for the ten evaluated configurations. Each panel is ranked independently by its own metric, so the reordering between panels shows that no single configuration leads on all three. The proposed model is highlighted in red.

ahead of the unmodified YOLOv8m at 0.8100 and 152.2 FPS. Two configurations achieve higher mAP@0.5 overall, the WBF ensemble at 0.8272 and RT-DETR at 0.8174, but each runs at a fraction of the throughput, and neither exceeds the proposed model at the fixed operating point.

The comparison against YOLOv8m in this table should be read cautiously, for the reason given in Section 6.6. The controlled comparison for the attention mechanism is the three-seed ablation of Section 7.3, not this table.

7.1.3 RT-DETR and the Calibration Effect

RT-DETR presents the clearest illustration of why two operating points are reported. It attains the second-highest mAP@0.5 at 0.8174 and by far the highest recall at 0.8523, yet the second-lowest F1 in the table at 0.6813, because its precision at the fixed threshold of 0.25 is only 0.5674. Swept to its own optimum at a confidence of 0.65, the same predictions yield the highest F1* of any configuration at 0.7919, an improvement of 11.1 percentage points over its fixed-threshold value.

Nothing about the model changed between those two numbers. Only the point at which its precision–recall curve was sampled. Set-prediction detectors emit a fixed-size set of queries without non-maximum suppression, and distribute confidence across that set differently from detectors whose outputs have been filtered by suppression. A threshold that is conventional for one family is therefore not neutral for another. Its throughput of 60.5 FPS is well below the YOLO-family single-model detectors, reflecting the cost of the transformer encoder and decoder.

7.1.4 The YOLO Family

YOLOv8m, YOLOv11m and YOLOv10m attain mAP@0.5 values of 0.8100, 0.7899 and 0.7598 at 152.2, 160.6 and 172.8 FPS respectively. The ordering is notable: YOLOv11m does not surpass YOLOv8m on this dataset despite being the more recent architecture, and YOLOv10m trades a further 3.0 percentage points of mAP@0.5 for approximately 8 percent more throughput. Pham et al. [Pham, Ngoc, and Bui 2024] report the same non-monotonicity across YOLO versions for road damage, and Section 7.8.2 argues that the YOLOv11m result here is not incidental.

7.1.5 Ensemble and Two-Stage Configurations

The WBF ensemble attains the highest mAP@0.5 at 0.8272 and the highest precision at 0.8564, but among the lowest recall at 0.6793. The mechanism is transparent: weighted box fusion retains detections supported by both member models with sufficient overlap, so agreement between members is rewarded and detections found by only one member are penalised. The result is a precision-oriented operating point with an F1 of 0.7576, below that of the proposed single model, at 23.6 FPS, roughly one sixth the throughput of the YOLO-family detectors. Faster R-CNN alone attains 0.7976 mAP@0.5 and 0.7365 F1 at 28.1 FPS.

7.1.6 Single-Attention Comparators

YOLOv8m+CBAM and YOLOv8m+CoordAttn attain mAP@0.5 values of 0.7782 and 0.7879 and F1 scores of 0.7424 and 0.7463, at 142.0 and 147.3 FPS. Both therefore perform comparably to, but do not improve upon, the unmodified YOLOv8m baseline when a single attention mechanism is applied under the residual-gated two-phase protocol. This is consistent with the ablation of Section 7.3, in which neither ECA nor CBAM applied alone exceeds the no-attention baseline, and it is the first indication that the effect reported for the proposed model belongs to the combination rather than to either component.

7.1.7 The Throughput-Oriented Lower Bound

SSD-VGG16 is the fastest configuration at 174.7 FPS but attains the lowest mAP@0.5 at 0.6165 and the lowest F1 at 0.5978. Its recall of 0.5253 indicates that it misses nearly half the annotated potholes. Two factors beyond architecture contribute: it operates at 300×300 rather than 640×640 , which is a substantial disadvantage for small objects, and it was trained for 20 epochs without validation-based checkpoint selection. Its result should therefore not be read as a clean statement about the SSD architecture. What it does establish is that high throughput alone does not imply usable pothole detection.

7.1.8 Error Modes Behind the Aggregate Metrics

The precision and recall figures of Table 7.1 compress detection counts that are worth examining directly. The validation partition contains 474 annotated instances. Three patterns emerge that the aggregate metrics obscure.

The configurations occupy distinct error regimes rather than differing in overall quality. RT-DETR misses only 70 of the 474 instances but emits 308 false positives. The WBF ensemble emits only 54 false positives but misses 152 instances. These are not a good and a bad detector but two detectors placed at opposite ends of the same trade-off, one by its NMS-free set-prediction design and the other by a fusion rule that requires agreement between members. Between them, YOLOv8m at 358 true positives and 102 false positives and RGDA-YOLOv8m at 363 and 109 occupy the balanced region.

The proposed model's improvement is visible in these counts as five additional correct detections and seven additional false alarms relative to YOLOv8m at seed 42. Stated that way the effect is unmistakably small, it concerns roughly one percent of the annotated instances, which is a useful corrective to reading 0.82 percentage points of mAP@0.5 as a substantial gain. It also clarifies why three seeds cannot resolve the effect size: a difference of five detections is comfortably within what a different random initialisation produces.

The false-negative counts are the ones that matter for the application. Every configuration misses between 15 and 47 percent of annotated potholes at the fixed threshold. Even the best-performing configurations therefore fail to report roughly one defect in four, which is the appropriate context for any claim about deployment readiness. The

Seed	mAP@0.5	mAP@0.5:0.95	Precision	Recall	F1
42	0.8160	0.5476	0.7691	0.7658	0.7674
123	0.8272	0.5414	0.8059	0.7637	0.7842
456	0.7990	0.5378	0.8017	0.7236	0.7607
Mean $\pm$	0.8141 $\pm$	0.5423 $\pm$	0.7922 $\pm$	0.7510 $\pm$	0.7708 $\pm$
std	0.0142	0.0050	0.0201	0.0238	0.0121
vs. base-line	+0.0082	—	+0.0023	+0.0175	+0.0107

Table 7.2: RGDA-YOLOv8m across three random seeds at confidence 0.25 and IoU 0.5. Deltas are relative to the three-seed unmodified YOLOv8m baseline of Table 7.3.

lowest false-negative count in the table belongs to RT-DETR, which is worth noting given that its fixed-threshold F1 is second-lowest: for an application in which a missed hazard is the costly error and a spurious alert merely annoying, RT-DETR at a properly calibrated threshold may well be the better choice than the model this dissertation proposes.

7.2 Multi-Seed Evaluation of RGDA-YOLOv8m

RGDA-YOLOv8m was trained independently at seeds 42, 123 and 456. Table 7.2 reports the individual and aggregated results. Deltas are computed against the three-seed unmodified YOLOv8m baseline of the ablation study (Table 7.3), which attains a mean mAP@0.5 of 0.8059 and a mean F1 of 0.7601.

Across the three seeds the model attains an mAP@0.5 of 0.8141 ± 0.0142 and an F1 of 0.7708 ± 0.0121 . Seed 123 produces the strongest run on both measures, at 0.8272 and 0.7842; seed 456 the weakest, at 0.7990 and 0.7607. The spread of 2.8 percentage points in mAP@0.5 between the best and worst runs of the *same configuration* is itself worth stating: it is three times the mean improvement being claimed, and it is the reason multi-seed reporting is not optional for effects of this size.

Relative to the three-seed baseline, mean mAP@0.5 improves by 0.82 percentage points and mean F1 by 1.07. The decomposition of the F1 gain matters: precision rises by 0.23 percentage points, from 0.7899 to 0.7922, and recall by 1.75, from 0.7335 to 0.7510. Both components improve, so the F1 gain is not a precision–recall trade being presented as an improvement: the model finds more potholes without producing proportionally more false alarms. Given the application, the recall-weighted character of the gain is the more useful direction.

Each of these mean improvements nonetheless lies within the corresponding seed-to-seed standard deviation. Taken alone, Table 7.2 does not establish the improvement. The paired evidence in Section 7.3 is what supports it.

Configuration	mAP@0.5	F1
YOLOv8m (no attention)	0.8059 ± 0.0107	0.7601 ± 0.0111
YOLOv8m + ECA only	0.7986 ± 0.0073	0.7533 ± 0.0068
YOLOv8m + CBAM only	0.8012 ± 0.0034	0.7544 ± 0.0079
RGDA-YOLOv8m (ECA+CBAM)	0.8141 ± 0.0142	0.7708 ± 0.0121

Table 7.3: Ablation of the residual-gated attention configurations across seeds 42, 123 and 456 (mean $\pm$ standard deviation, confidence 0.25, IoU 0.5). All four conditions share the partition, injection points, gate and training schedule.

7.3 Ablation Study

7.3.1 Results

Table 7.3 reports the four cells of the factorial design of Section 5.8, each at three seeds under an identical training schedule.

Neither single-attention configuration improves on the no-attention baseline in the mean. ECA alone reduces mean mAP@0.5 by 0.73 percentage points and mean F1 by 0.68. CBAM alone reduces them by 0.47 and 0.57 respectively. In both cases the mean decrease is smaller than the seed-to-seed standard deviation of the variant concerned, so these are not confirmed degradations: the honest statement is that neither single module produces a detectable improvement.

Only the combined configuration exceeds the baseline, at 0.8141 mAP@0.5 and 0.7708 F1, corresponding to improvements of 0.82 and 1.07 percentage points.

7.3.2 Interpretation: A Non-Additive Interaction

An additive model of attention predicts that if ECA contributes δ_E and CBAM contributes δ_C , their combination contributes approximately $\delta_E + \delta_C$. Here $\delta_E = -0.73$ and $\delta_C = -0.47$ percentage points of mAP@0.5, so an additive account predicts approximately -1.20 for the combination. The measured value is $+0.82$, a discrepancy of 2.0 percentage points in the opposite direction. Whatever the combined block is doing, it is not the sum of what its parts do.

This is directionally consistent with the hypothesis of Section 5.1: that the two mechanisms address different failure modes and that each is insufficient alone. Suppressing an uninformative channel helps only if the spatially relevant region is also emphasised. Emphasising a region helps only if the channels describing it have been selected. On this reading the combination is not two improvements stacked but one mechanism completed.

The evidence supports the direction of that claim more strongly than its magnitude. The improvement of the combined configuration does not exceed its own seed-to-seed standard deviation of ± 0.0142 in mAP@0.5 and ± 0.0121 in F1. What is better supported is the paired comparison: across the three seeds the combined configuration exceeds

the identically trained baseline in *every* run, by 0.62, 1.30 and 0.52 percentage points of mAP@0.5. Sign consistency across all paired runs is stronger evidence than the overlapping error bars suggest, because the pairing removes the seed-level variation common to both configurations. It remains true that three paired observations cannot establish an effect size.

7.3.3 Reconciling the Two CBAM Results

Table 7.1 reports YOLOv8m+CBAM at an mAP@0.5 of 0.7782, while Table 7.3 reports the CBAM-only condition at a mean of 0.8012. The gap of 2.3 percentage points requires explanation, since both use the same zero-initialised gate, the same three injection points and the same two-phase schedule.

The difference is experimental rather than architectural. The benchmark entry is a single training run conducted as part of a separate experiment. The ablation entry is a mean over three seeds within one experiment. The gap is comparable to the seed-to-seed dispersion observed across the ablation configurations, which ranges from ± 0.0034 to ± 0.0142 in mAP@0.5, and is therefore attributable to run-to-run variation. It is retained in the text rather than reconciled away because it is a concrete instance of the general warning in Section 6.6: a single-run number for a configuration can differ from its own three-seed mean by more than the effect under study.

7.4 Architecture Dependence of External Attention

7.4.1 The Experiment

The ablation establishes that residual-gated ECA+CBAM improves on the no-attention baseline when the host is YOLOv8m. Whether that finding is a property of the block itself is a separate question. YOLOv11m provides the test case, because its backbone contains a C2PSA partial self-attention module after the spatial pyramid pooling stage, so the features reaching its neck have already undergone one round of content-dependent spatial reweighting, whereas those of YOLOv8m have not.

The identical block (same modules, same gate, same layer-selection rule, same two-phase schedule, same three seeds) was attached at the three corresponding YOLOv11m neck stages. Because the injector selects layers positionally by class name across the set {C3k2, C2f, C2fAttn, RepC3, C3}, no modification of any kind was required to move it between hosts, which is what makes the comparison controlled.

7.4.2 Results

Adding the block to YOLOv11m reduced mean mAP@0.5 from 0.7899 to 0.7823 ± 0.0048 and mean F1 from 0.7348 to 0.7266 ± 0.0040 . The per-seed mAP@0.5 values were 0.7777, 0.7873 and 0.7818, all below the 0.7899 baseline. The per-seed F1 values were 0.7290, 0.7220 and 0.7288, all below the 0.7348 baseline. The degradation therefore held in

Host	Configuration	mAP@0.5	F1
YOLOv8m	no attention	0.8059 ± 0.0107	0.7601 ± 0.0111
	+ ECA+CBAM	0.8141 ± 0.0142	0.7708 ± 0.0121
	<i>effect</i>	+0.0082	+0.0107
YOLOv11m	no attention	0.7899	0.7348
	+ ECA+CBAM	0.7823 ± 0.0048	0.7266 ± 0.0040
	<i>effect</i>	-0.0076	-0.0082

Table 7.4: Effect of attaching the identical residual-gated ECA+CBAM block to two host architectures across seeds 42, 123 and 456. The effect reverses in sign.

the same direction at every seed and, at 0.76 and 0.82 percentage points, exceeded the seed-to-seed standard deviation of the attention arm itself, which is 0.48 and 0.40 percentage points in the two metrics.

One asymmetry must be stated. The YOLOv11m baseline of 0.7899 is a single training run, conducted as part of the cross-architecture benchmark rather than alongside the attention runs, so the comparison is between a three-seed distribution and a point estimate rather than a paired one. The YOLOv8m comparison of Section 7.3 does not share this weakness, because both of its arms were trained at three seeds within one experiment.

7.4.3 What This Result Does and Does Not Establish

The experiment’s strength is that it holds constant everything except the variable of interest. The attention block, the gate, the insertion rule, the training schedule, the dataset partition, the seeds and the evaluation protocol are all identical between the two arms. The host architecture is the only thing that changes, and the sign of the effect changes with it.

Its weakness is the single-run baseline. Section 7.3 reconciled a 2.3 percentage-point gap between a single-run and a three-seed figure for the same CBAM configuration, and that gap is three times the 0.76 percentage points separating the two YOLOv11m arms here. The reversal is therefore consistent in sign across three seeds but not established at the strength the YOLOv8m ablation reaches, and confirming it would require training the stock YOLOv11m baseline at the same three seeds.

Read with that qualification, the implication still applies beyond this dissertation: a report that an attention module improves one detector is not by itself evidence that it will improve another, even a close relative in the same architectural family. Section 7.8.2 develops this.

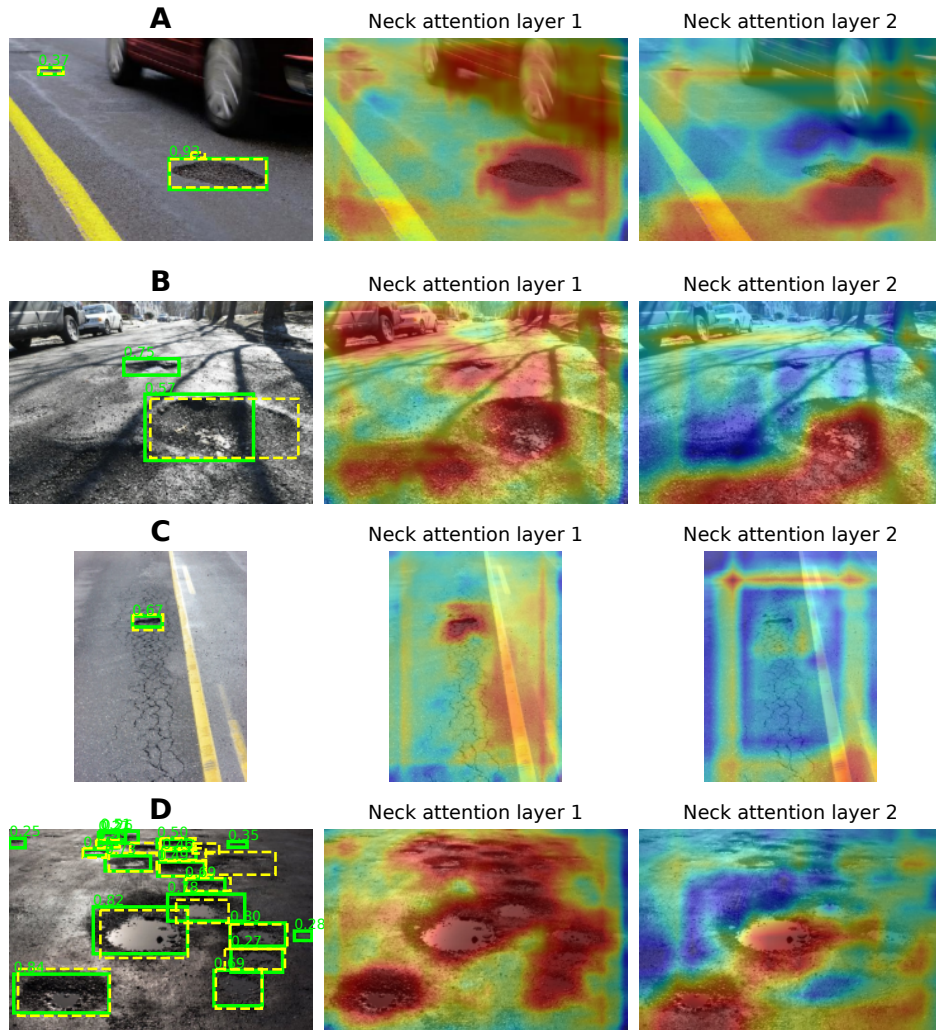


Figure 7.2: CBAM spatial-attention maps from the seed-123 RGDA-YOLOv8m model on four validation images. Left: predictions (solid green) and annotations (dashed yellow). Centre and right: spatial attention at the middle ($C = 384$) and deep ($C = 576$) injection points.

7.5 Qualitative Analysis

Figure 7.2 shows CBAM spatial-attention maps produced by the seed-123 model for four validation images, covering potholes on gravel, wet tarmac, dry asphalt and a coloured road surface. The first column shows predictions in solid green against annotations in dashed yellow. The second and third show the spatial-attention maps at the middle and deep injection points, where $C = 384$ and $C = 576$, with warmer colours indicating larger weights.

Several high-attention regions coincide with annotated potholes, and the deep map

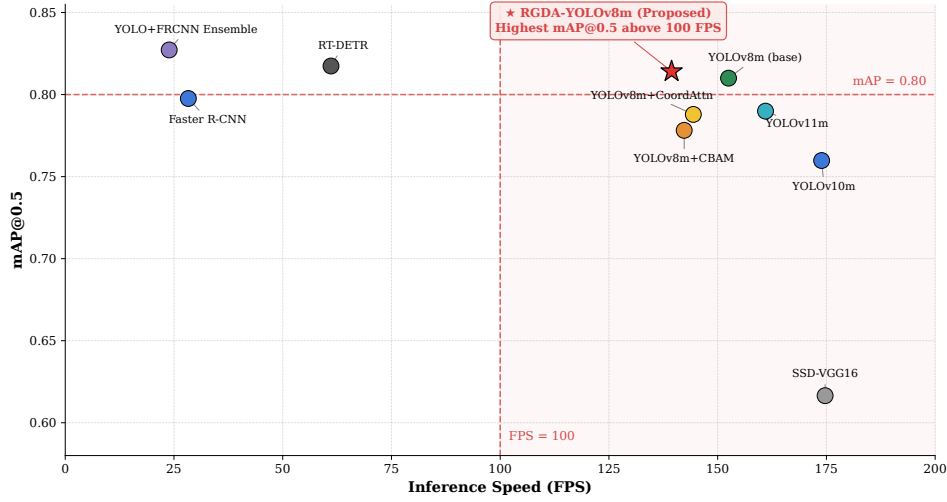


Figure 7.3: Inference throughput and mAP@0.5 of the ten evaluated configurations. The shaded region marks the preferred operating area above 0.80 mAP@0.5 and 100 FPS.

is visibly coarser than the middle one, consistent with its lower spatial resolution. Two cautions apply to reading these maps. They visualise $\mathbf{M}_s$ in isolation, whereas the block's actual contribution to the output is scaled by $\tanh(\alpha)$ and follows channel recalibration, so a strong spatial weight does not by itself indicate a strong effect on the prediction. And correspondence between a highlighted region and an annotation is consistent with the mechanism working as intended but does not demonstrate that the highlighting caused the detection. These maps are presented as qualitative support, not as evidence.

7.6 Speed–Accuracy Trade-Off

Figure 7.3 plots mAP@0.5 against throughput for all ten configurations. Three groups emerge.

Accuracy-oriented. The two highest mAP@0.5 values belong to the two slowest configurations: the WBF ensemble at 0.8272 and 23.6 FPS, and RT-DETR at 0.8174 and 60.5 FPS. Faster R-CNN at 0.7976 and 28.1 FPS is strictly dominated by RT-DETR, which is both more accurate and more than twice as fast. Accuracy in this group costs a factor of two to seven in throughput relative to the single-model YOLO detectors.

Balanced. RGDA-YOLOv8m attains 0.8141 mAP@0.5 at 139.9 FPS, the highest accuracy of any configuration above 100 FPS. Against the unmodified YOLOv8m of Table 7.1, the attention mechanism costs 8.1 percent of throughput. The exchange is therefore 8.1 percent of speed and 0.25 percent of parameters for 0.82 percentage points of mAP@0.5 and 1.07 of F1, an exchange whose attractiveness depends entirely on whether the application has throughput headroom, which at 139 FPS on this hardware it clearly

does.

Throughput-oriented. SSD-VGG16 and YOLOv10m reach 174.7 and 172.8 FPS at mAP@0.5 of 0.6165 and 0.7598. YOLOv10m represents a defensible operating point where throughput is paramount. SSD-VGG16 does not, since its recall of 0.5253 means it misses nearly half of all annotated instances.

Only three of the ten configurations are strictly dominated: Faster R-CNN, less accurate and slower than RT-DETR; and YOLOv8m+CBAM and YOLOv8m+CoordAttn, both less accurate and slower than the unmodified YOLOv8m. That the two single-attention variants are dominated by the model they modify is a compact statement of the ablation result. The remaining seven occupy distinct positions on the frontier, so the choice among them is determined by the throughput budget of the target application rather than by a single dominant configuration.

All figures were measured on an RTX 3090 at FP32 and batch size one. They establish real-time performance on desktop-class hardware and do not extend to embedded or quantised deployment.

7.7 Comparison With Published Studies

Table 7.5 places these results alongside representative published work. Because those studies use different datasets, defect classes, hardware and evaluation protocols, the comparison provides context rather than a ranking, a caveat that Chapter 2 argued applies to the entire field.

RGDA-YOLOv8m's mAP@0.5 of 0.8141 is approximately 1.1 percentage points below the 0.825 reported by Paramarthalingam et al. [Paramarthalingam et al. 2024] while operating at roughly 4.7 times their throughput, so the contribution relative to that work is speed at comparable accuracy rather than higher accuracy. Its F1 of 0.7708 far exceeds the SSD-MobileNetV2 result of 0.479 reported by Asad et al. [Asad et al. 2022] but falls below their YOLOv5 F1 of 0.870.

That last gap deserves a candid reading rather than a defensive one. It is consistent with the corpus used here being smaller and, critically, deduplicated. Chapter 4 showed that more than half the images in the union of three Kaggle pothole repositories are duplicates. A study drawing on such sources without deduplication will have near-identical images in both partitions and will report inflated metrics. Since the studies in Table 7.5 do not report a deduplication step, the possibility that part of the gap reflects evaluation conditions rather than detector quality cannot be excluded, and, equally, cannot be demonstrated. The honest conclusion is that these numbers are not comparable, which is the point Chapter 2 made about the field as a whole.

Study	Method	Dataset	Reported	FPS
Paramarthalingam et al. [2024]	YOLOv5	Kaggle pot-hole	mAP@0.5 = 0.825; P = 0.862; R = 0.759	30
Asad et al. [2022]	YOLOv5 vs. SSD-MobileNetV2	Kaggle pot-hole	F1 = 0.870 (YOLOv5); 0.479 (SSD)	—
Al-Shaghouri, Alkhatib, and Berjaoui [2021]	YOLOv4-CSPDarknet53	1,087 pothole images	mAP = 0.854; P = 0.85; R = 0.81	20
Ding et al. [2022]	One- and two-stage ensemble	RDD2022 / CRDDC	Mean F1 = 0.770 over three countries	—
Pan, Xu Wang, and Huan [2025]	RT-DSAFDet	UAV-PDD2023	mAP@0.5 = 0.542	—
Abdelwahed et al. [2025]	YOLOv10m / v11m / v12m	RDD (T4, TensorRT)	mAP@0.5:0.95 = 0.511 / 0.515 / 0.525	223 / 213 / 206
This work (2026)	RGDA-YOLOv8m	926-image deduplicated	mAP@0.5 = 0.814; F1 = 0.771	140

Table 7.5: Representative published pothole and road-damage detection results. Datasets, classes, hardware and protocols differ, so these values provide context rather than a controlled ranking.

7.8 Discussion

7.8.1 What the Ablation Shows

The pattern in Table 7.3 is not what a purely additive account of attention predicts. Applied alone, ECA reduces mean mAP@0.5 by 0.73 percentage points and CBAM by 0.47, yet their sequential combination increases it by 0.82. Neither module is individually beneficial on this data, so the gain of the combined block cannot be attributed to either pathway.

The reading offered here is that the two address different and jointly necessary failure modes. Potholes are low-contrast regions whose appearance overlaps with shadows, patched asphalt and wet surfaces, so suppressing an uninformative channel helps only if the spatially relevant region is also emphasised, and emphasising a region helps only if the channels describing it have been selected. Because CBAM already contains a channel branch (Equation (5.5)), the ECA factor in this design measures the contribution of an additional, parameter-light channel interaction placed before spatial refinement, not the contribution of channel attention as such, a distinction that changes how the ablation should be read.

This interpretation is consistent with the measurements rather than demonstrated by them, and the distinction is worth preserving. What can be stated with more confidence is the paired result: across three seeds the combined configuration beats the identically trained baseline in every run. What cannot be stated is the size of the effect.

7.8.2 Attention Saturation

The YOLOv11m experiment reverses the sign of the effect: attaching the identical block reduced mean mAP@0.5 from 0.7899 to 0.7823 and mean F1 from 0.7348 to 0.7266, in the same direction at every seed and by more than the attention arm’s own seed-to-seed variation. As Section 7.4 notes, the baseline is a single run, so the reversal is suggestive rather than established.

The two hosts differ in what they already provide. YOLOv11m places a C2PSA partial self-attention module after the spatial pyramid pooling stage of its backbone, so features reaching the neck have already undergone one round of content-dependent spatial reweighting. YOLOv8m has no such stage. The results are consistent with the view that the benefit of an added attention block depends on whether the host has left the corresponding recalibration undone, and that stacking a second mechanism onto already-reweighted features can be actively harmful rather than merely redundant. The term used here for that condition is *attention saturation*.

Two mechanisms could produce the observed harm, and this study does not distinguish between them. The first is representational: a second reweighting of already-reweighted features compounds the suppression, so features attenuated by C2PSA are attenuated again and useful signal is lost. The second is optimisational: the gradient reaching the backbone now passes through two sequential gating nonlinearities, which may degrade the fine-tuning of the pretrained weights on a small dataset. Distinguishing them would require measuring the learned α values and the feature statistics on both hosts, which is proposed as future work in Section 8.5.

This finding stands in apparent tension with Zhang et al. [Linxuan Zhang, Deng, and Zou 2025], who report that several attention modules added to a YOLOv11 host improve mAP@0.50 on RDD2022, coordinate attention by approximately 1.6 percentage points. The two results are not necessarily contradictory. Their modifications also replace C3k2 blocks and restructure multi-scale feature extraction, so the attention is added to an architecture that has itself been altered rather than to a stock YOLOv11 neck. Their insertion points, module types, dataset, class taxonomy and training schedule all differ; and RDD2022 is far larger than the corpus used here, which changes how much perturbation fine-tuning can absorb. What the two results jointly establish is precisely the claim being made: the effect of an attention module is contingent on its context, and neither result generalises to the other’s setting. The observation of Youwai et al. [Youwai, Chaiyaphat, and Chaipetch 2024] that a single-attention-step variant underperformed their multi-step configuration points in the same direction: the arrangement and quantity of attention matters independently of its presence.

The practical consequence for the literature is uncomfortable but simple. Chapter 2

found that attention modules are almost uniformly reported as beneficial in the road-damage literature, and that those reports rest on single-host, single-dataset experiments. If the effect can reverse in sign between two members of the same architectural family, as it does here between YOLOv8m and YOLOv11m, then such reports do not support the general claim they are routinely read as making.

7.8.3 Protocol Sensitivity

Three methodological choices made during this work each changed measured outcomes by more than the architectural effect the work set out to detect.

Deduplication removed 1,078 of 2,004 images, more than half the corpus, because two of the three source repositories redistribute overlapping data. Without it, near-duplicate images appear in both partitions and every reported figure is inflated. A related error (baselines inadvertently scored over the full un-deduplicated corpus, producing a Faster R-CNN recall of 0.9988) was caught only because the implied ground-truth count did not match the partition being scored.

The average-precision definition is not interchangeable. Scoring the same YOLOv11m+ECA+CBAM predictions with the PASCAL VOC 2007 11-point rule and with COCO-style 101-point interpolation gives mean mAP@0.5 of 0.7555 and 0.7823 respectively, a difference of 2.7 percentage points, more than three times the improvement attributed to the proposed block.

Throughput measurement is equally sensitive. Timing inference inside an evaluation loop, where image decoding is interleaved with forward passes, understated throughput by 20 to 26 percent relative to a dedicated pass with pre-decoded images, and inverted the relative ordering of the YOLO variants.

None of these choices is novel or contentious once stated. Their significance is that each was capable of producing an apparent effect of the same magnitude as the architectural contribution under investigation. This argues for reporting evaluation protocol in enough detail for a reader to determine whether two sets of numbers are comparable at all, and it suggests that some fraction of the improvements reported in this literature are protocol artefacts that no reader can currently identify as such.

7.8.4 Limitations

Statistical power. Three seeds establish that an effect is consistent in sign but cannot bound its magnitude, and the mean improvement of the proposed configuration lies within its own seed-to-seed standard deviation. A larger seed budget, or a validation partition larger than 186 images and 474 instances, would be required to characterise the effect with confidence. This is the most serious limitation of the positive result.

The YOLOv11m baseline is a single run. The attention arm of the architecture-dependence experiment was trained at three seeds but the stock YOLOv11m comparator was not, so that comparison is unpaired and its 0.76 percentage-point effect is smaller than the single-run-to-three-seed discrepancy documented in Section 7.3. The reversal

is consistent across seeds but is not established as firmly as the YOLOv8m result.

Single-run comparators. The configurations in Table 7.1 are single training runs conducted as separate experiments, whereas the proposed model is a three-seed mean. The reconciliation of the two CBAM figures in Section 7.3 demonstrates the size of the resulting uncertainty directly.

Unmatched training budgets. Faster R-CNN and SSD-VGG16 were trained for 20 epochs at batch sizes of 4 and 1, with final-epoch weights retained and no validation-based checkpoint selection, whereas the YOLO-family models were trained for 50 epochs with validation-selected checkpoints. SSD-VGG16 additionally operates at 300×300 . Its weak result should not be attributed to the architecture alone.

Validation set reused for selection. With 926 images no separate test partition was held out, so the validation set serves both for checkpoint selection and for reporting in the YOLO-family models. This introduces an optimistic bias that applies to all such configurations equally.

Hook-based injection. The attention modules are attached through forward hooks rather than registered as submodules, so the injected parameters are absent from the standard checkpoint and must be persisted separately. A model distributed as weights alone silently reverts to an unmodified detector.

Scope. The evaluation covers one deduplicated dataset with one object class, in predominantly daylight and fair-weather conditions, from ground-level viewpoints. All throughput measurements used an RTX 3090 at FP32 and batch size one. The results do not extend to embedded or quantised deployment, to video sequences where temporal consistency matters, or to the adverse conditions studied by Frnda et al. [Frnda et al. 2024].

Chapter 8

Conclusion

8.1 Summary of Contributions

This dissertation used the Ultralytics framework and a purpose-built experimental pipeline to conduct a controlled comparison of ten detector configurations on a single pothole corpus, with a residual-gated dual-attention block as the focal architecture. The corpus was assembled from three public Kaggle repositories and deduplicated by normalized-pixel nearest-neighbor matching, which removed 1,078 of 2,004 images and left 926 images carrying 2,354 annotations. Every configuration was fine-tuned from a COCO-pretrained initialisation on the same 740-image training partition, scored on the same 186-image validation partition under one average-precision definition, and timed under one throughput procedure, so that differences between them reflect the detectors rather than the measurement.

The study contributes more than a comparison. It introduces RGDA-YOLOv8m, which applies Efficient Channel Attention followed by the Convolutional Block Attention Module at the three multiscale fusion stages of the YOLOv8m neck and combines the refined tensor with the unrefined one through a learnable scalar gate initialised to zero, so that the modified detector begins each training phase as an exact copy of the model it inherits. It isolates the individual and joint contributions of the two attention pathways through a factorial ablation run at three seeds under one schedule. And it asks whether the resulting benefit is a property of the block or of the block–host pairing, by attaching the identical mechanism to YOLOv11m, whose backbone already performs one round of spatial self-attention before its neck begins fusing scales.

8.2 Key Findings

The findings of this dissertation, against the objectives of Section 1.2, are as follows.

Cross-architecture comparison. No configuration leads on all measures. The WBF ensemble attains the highest mAP@0.5 at 0.8272 but at 23.6 FPS. RT-DETR attains the highest F1* at 0.7919 but at 60.5 FPS. RGDA-YOLOv8m attains the highest fixed-

threshold F1 at 0.7708 while running at 139.9 FPS. Among configurations exceeding 100 FPS, the proposed model attains the highest mAP@0.5. Notably, YOLOv11m did not surpass YOLOv8m on this corpus, and SSD-VGG16 demonstrated that high throughput alone does not imply usable detection.

The gated dual-attention block on YOLOv8m. It helps, by a small margin. Across three seeds it attained 0.8141 ± 0.0142 mAP@0.5 and 0.7708 ± 0.0121 F1, improvements of 0.82 and 1.07 percentage points over the identically trained baseline, with both precision and recall rising so that the F1 gain is not a trade. The improvement was consistent in sign across all three paired runs, at 0.62, 1.30 and 0.52 percentage points of mAP@0.5, but its mean lies within its own seed-to-seed standard deviation. The correct characterisation is a consistent small improvement of unestablished magnitude.

Independence of the two pathways. The pathways do not contribute independently. ECA alone reduced mean mAP@0.5 by 0.73 percentage points and CBAM alone by 0.47, while their combination increased it by 0.82. An additive account predicts approximately -1.20 . The measured value differs from that prediction by 2.0 percentage points in the opposite direction. The effect belongs to the combination.

Transfer to YOLOv11m. The benefit does not transfer. It reverses. Attaching the identical block to YOLOv11m reduced mean mAP@0.5 from 0.7899 to 0.7823 ± 0.0048 and mean F1 from 0.7348 to 0.7266 ± 0.0040 , at every seed. The baseline is a single run rather than a three-seed mean, so the reversal is consistent in sign but less firmly established than the YOLOv8m improvement. It is nonetheless the finding with the broadest implication.

8.3 Comparative Insights

The benchmark showed that the ranking of detectors on this task depends on which question is asked. The Weighted Box Fusion ensemble attained the highest mAP@0.5 at 0.8272 but ran at 23.6 FPS, RT-DETR attained the highest swept F1 at 0.7919 but at 60.5 FPS, and RGDA-YOLOv8m attained the highest fixed-threshold F1 at 0.7708 while sustaining 139.9 FPS. Only three of the ten configurations were strictly dominated, so the choice among the remaining seven is settled by the throughput budget of the target application rather than by a single best model.

The detection counts behind those aggregates were more informative than the aggregates themselves. RT-DETR missed only 70 of the 474 annotated instances but emitted 308 false positives, while the ensemble emitted only 54 false positives and missed 152 instances. These are not a better and a worse detector but two positions on one trade-off, the first produced by NMS-free set prediction and the second by a fusion rule that rewards agreement between members. RGDA-YOLOv8m, at 363 true positives against 109 false positives, occupied the balanced region between them.

RT-DETR also showed how much of a reported F1 can be an artefact of threshold placement rather than of detection capability. At the fixed confidence of 0.25 its F1 was 0.6813, the second lowest in the benchmark, and swept to its own optimum at 0.65 the same predictions yielded 0.7919, the highest of any configuration. The 11.1 percentage

points between those two numbers exceeded the entire spread of the remaining nine models, which is why two operating points were reported throughout rather than one.

Within the YOLO family, accuracy and recency did not move together. YOLOv8m, YOLOv11m and YOLOv10m attained 0.8100, 0.7899 and 0.7598 mAP@0.5 at 152.2, 160.6 and 172.8 FPS, so on this corpus the more recent architectures were the faster and the less accurate. Pham et al. [Pham, Ngoc, and Bui 2024] report the same non-monotonicity for road damage, and Section 7.8.2 argues that the position of YOLOv11m in that ordering is connected to the architecture-dependence result rather than incidental to it.

8.4 Limitations and Trade-offs

The central limitation concerns statistical power rather than the mechanism. The improvement of 0.82 percentage points in mAP@0.5 lies inside the seed-to-seed standard deviation of the proposed configuration itself, which is 1.42 percentage points, and the spread between its best and worst seeds was 2.8 percentage points, more than three times the effect being claimed. What three seeds establish is that the improvement is consistent in sign across every paired run. What they cannot establish is its magnitude. The same shortage of runs weakens the architecture-dependence result further, because the YOLOv11m baseline was a single training run rather than a three-seed mean, so that effect too is signed but not sized. Section 7.8.4 sets out the remaining experimental asymmetries in full.

The accuracy gain was bought with throughput. The attention block cost 8.1 percent of the unmodified detector's frame rate in exchange for 0.25 percent additional parameters, and the attractiveness of that exchange depends entirely on whether the deployment has headroom to spend. At 139.9 FPS on desktop-class hardware it plainly does. On an embedded accelerator running at a fraction of that rate it may not.

A final trade-off belongs to the application rather than to the models. Every configuration missed between 15 and 47 percent of annotated potholes at the fixed operating threshold, so even the best of them failed to report roughly one defect in four. For an assistive system in which a missed hazard is the costly error and a spurious alert merely an annoyance, RT-DETR at a properly calibrated threshold may well be the better choice than the model proposed here, despite its lower fixed-threshold F1.

8.5 Future Directions

8.5.1 Diagnosing Attention Saturation

Section 7.8.2 identified two candidate mechanisms for the YOLOv11m degradation, representational over-suppression and optimisation difficulty through stacked gating nonlinearities, and did not distinguish them. Recording the learned α values per injection point on both hosts would be directly informative: if the YOLOv11m gates

converge towards zero the network is declining the pathway, whereas if they open and performance still falls the mechanism is representational. Comparing the channel and spatial statistics of neck features before and after C2PSA on the two hosts would test the over-suppression account directly.

The broader goal is a *predictive* account. If saturation can be diagnosed from properties of the host (the presence and position of existing attention stages, or measurable statistics of the features arriving at the injection point) then whether to add an attention module becomes a decision that can be made by inspection rather than by training. That would be of considerably more practical value than any individual architectural result in this dissertation.

8.5.2 Architectural and Training Refinements

Several design decisions were fixed rather than explored. A per-scale analysis of the three learned gates would reveal whether attention is uniformly useful across scales or concentrated at one. If concentrated, the mechanism could be pruned at essentially no accuracy cost. Carrying the Phase-1 attention module into Phase 2, rather than discarding it as the present protocol requires, would test whether a genuinely two-stage attention curriculum outperforms the current domain-adaptation-then-attention arrangement. A per-channel rather than scalar gate, and a parallel rather than sequential arrangement of the two mechanisms, are both worth evaluating. Registering the attention modules as proper submodules, accepting a modified model definition in exchange, would eliminate the checkpoint-persistence failure mode entirely.

8.5.3 Deployment and Robustness

The natural extension is deployment. Quantisation to INT8 and export through TensorRT to embedded accelerators such as the Jetson family would establish whether the accuracy-throughput exchange survives at the operating points where throughput is actually scarce, and whether the attention modules quantise gracefully. Extension from images to video would allow temporal consistency to be exploited, since a pothole detected in one frame constrains where it should appear in the next, and would permit evaluation by track-level rather than frame-level metrics, which is closer to what both target applications actually require.

Robustness is the largest gap. The corpus contains essentially no night, rain, snow or low-sun imagery, and Frnda et al. [Frnda et al. 2024] show that detection accuracy degrades sharply and unevenly under exactly those conditions. Evaluation on a deliberately adverse-condition corpus is a prerequisite for any safety claim.

A more fundamental answer to both the condition and the viewpoint limitation is to change the sensing rather than the corpus. An unmanned aerial platform carrying registered RGB, LiDAR and thermal channels would address three of this dissertation's constraints at once. The aerial viewpoint covers the ground-level-only bias recorded in Section 4.6, and UAV-PDD2023 [Pan, Xu Wang, and Huan 2025] establishes that the

perspective is workable, although the mAP@0.5 of 0.542 reported on it indicates that it is also harder. LiDAR measures the property that actually defines the defect. A depression remains a depression whether it is shadowed, wet or freshly patched, so the confounders identified in Section 1.1.1 largely dissolve under a geometric measurement, which is the argument Fan et al. [Fan et al. 2022] make for reconstruction-based detection. Thermal imaging supplies a third, orthogonal cue, since retained moisture and disturbed aggregate alter surface emissivity and cooling rate, and it supplies it at night, the condition under which Frnda et al. [Frnda et al. 2024] show RGB detection degrading most sharply.

Two obstacles should be stated alongside the proposal. No registered multi-modal pothole corpus of usable size exists, so the data would have to be collected rather than assembled from public repositories, and Ma et al. [N. Ma et al. 2022] note that sensing-hardware cost is precisely what has kept 3-D methods out of consumer-scale deployment, an objection that applies with more force to a three-sensor airborne platform than to a stereo rig. The extension would also change the question this dissertation asked. Fusing modalities calls for cross-modal attention rather than the within-tensor recalibration studied here, so neither the ablation result nor the saturation result of Section 7.8.2 would carry over, and both would have to be re-established in that setting.

8.5.4 Broader Evaluation

Extending the evaluation to multi-class road damage on RDD2022 [Arya et al. 2022] would test whether the ablation and saturation results hold when the detector must discriminate between damage types rather than detect a single class, and would place the work on a benchmark where cross-study comparison is at least possible. Integration into an end-to-end assistive navigation system (detection, tracking, distance estimation and alerting) would surface the requirements that a detection-only evaluation cannot, in particular the relationship between detection range and useful warning time identified by Al-Shaghouri et al. [Al-Shaghouri, Alkhatib, and Berjaoui 2021].

8.5.5 Summary

In summary, a diagnostic account of attention saturation, per-scale and per-channel refinements of the gate, multi-sensor aerial acquisition, and evaluation on multi-class road damage represent natural next steps that build directly on the findings of this dissertation. Each targets a limitation observed in the experiments: a reversal whose mechanism was not identified, a gate whose learned value was never inspected, a corpus containing neither adverse conditions nor aerial viewpoints, and a single-class task on which cross-study comparison remains impossible. Together they outline a path from a controlled architectural result to a detector that could be deployed and trusted.

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